SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE MEAN SQUARE OF THE CUSPIDAL ERROR OF $\theta^2(mz)\theta^2(m'z)$, ALMOST ALL N, AND THE SIZE OF THE PETERSSON NORM

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ABSTRACT. Let m, m' be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m,m';N) - M(m,m';N)$ is the error term in the number of representations of N by $m(x^2+y^2)+m'(z^2+w^2)$. Part I proved $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for the Petersson norm and deduced the level of distribution $1/2$ for the weighted sequence $r(n)r(N-n)$ and the bounds $\Omega(n(N-n)) \le 8$, $\max(\Omega(n), \Omega(N-n)) \le 5$ for every large $N \equiv 2 \pmod 4$. Here we prove a mean-square bound $\sum_{N \le Y} |a_G(N)|^2 \le 4\pi e^2 Y(1 + 54Y/L)\|G\|^2$ for every $G \in S_2(\Gamma_0(L))$ by a covering argument; for $G_{m,m'}$ it says that the root mean square of $(T-M)/\sqrt{N}$ is $\ll (mm')^{-1/2+\varepsilon}$, and it gives the level of distribution $1-\varepsilon$ for the weighted sequence for all but $O(X^{1-\delta})$ integers $N \le X$, and for those N the bounds 5 and 4 (a result which, as an arithmetic statement about almost all N, is superseded by the circle method, which gives $N = p + q$ with $p \equiv q \equiv 1 \pmod 4$). We reformulate the uniform bound of Part I in terms of Poincaré series, state the individual bound $|T - M| \ll N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}$ suggested by the data, and show that any power saving in mm' for individual N raises the level of distribution for every N beyond $1/2$. Finally we prove that the norm depends only on mm' (an Atkin–Lehner involution maps $G_{m,m'}$ to $G_{1,mm'}$), the lower bound $\|G_{m,m'}\|^2 \ge \frac{1}{\pi}(\log mm')^2 - O\big(\log(2mm')(\log\log 3mm')^C\big)$, by integrating $|G|^2$ over the parts of the fundamental domain above the Atkin–Lehner cusps at which $G_{m,m'}$ is transformed into $G_{1,mm'}$, whose first mm' coefficients are $r(n) - M(1, mm'; n)$, and the matching upper bound $\|G_{m,m'}\|^2 \ll (\log 3mm')^2(\log\log 3mm')^C$, so that the norm is of the exact order $(\log mm')^2$ when mm' has boundedly many prime factors and the uniform bound for $T - M$ holds with $(\log 3mm')(\log\log 3mm')^C$ in place of $(mm')^\varepsilon$. By an explicit Rankin–Selberg unfolding, $\langle \Theta_{m,m'}, f(d\,\cdot)\rangle = \pm\frac{4}{\pi^2}\prod_{p \mid mm'/M_1}(1+\chi_{-4}(p))\, d^{-1}L(1,f)L(1, f\otimes\chi_{-4})$ for every newform f of level $2^e M_1 \mid 4mm'$ and every d, with no other local factors, so these bounds are bounds for the second moment of $L(\frac12, f)L(\frac12, f\otimes\chi_{-4})$ over the family. Numerically, the regions above all the cusps account for 93–98% of the norm; there is an exact check at the level 20, and the supremum of $|T-M|/\sqrt{N}$ over N exceeds the root mean square by a bounded factor at every level computed. Finally, replacing the vector sieve of Part I by the two-dimensional sieve of Diamond, Halberstam and Richert, whose functions F_2, f_2 we compute, gives $\Omega(n(N-n)) \le 7$ with $\max(\Omega(n), \Omega(N-n)) \le 5$ for every large $N \equiv 2 \pmod 4$ from the same level $1/2$; the two inequalities for F_2, f_2 that this needs are certified in interval arithmetic.

1. RESULTS

We keep the notation of [5], referred to as Part I. Thus $r(n)$ is the number of representations of n as a sum of two squares, m, m' are coprime odd positive integers,

$$T(m,m';N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N\ge 0} T(m,m';N)e(Nz) \in M_2(\Gamma_0(L)),$$

Date: October 24, 2026 (draft).

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m,m';N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m,m';N) - M(m,m';N) \qquad \text{for every } N \geq 1.$$

The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$. Lemma 2.1 of Part I, which is Waibel's theorem [17, Theorem 1] for the form $Q_{m,m'} = m(x^2+y^2) + m'(z^2+w^2)$, gives $\|G\| \ll (mm')^{1/4+\varepsilon}$, and the first two versions of Part I deduced from it, through a Hecke operator, the uniform bound $a_G(N) \ll d(N) N^{1/2} (mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \leq 1$, the level of distribution $2/5$ and the bounds 9 and 6, and conjectured (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. Version 3 of Part I, which this paper accompanies, proves that the norm itself is much smaller than Waibel's general bound, by computing the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ with Waibel's method, and draws the consequences. We recall what we use.

Theorem 1.1 (Part I, Theorems 1.2 and 2.3)**.** *Let m, m' be coprime odd positive integers.*

(i) $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$.

(ii) *For $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$,*

$$T(m,m';N) = M(m,m';N) + O_\varepsilon\big(d(N)\, N^{1/2} (mm')^\varepsilon\big);$$

for cube-free m, m' and every N with $v_2(N) \leq 1$ the error term is $O_\varepsilon(d(N)^4 N^{1/2} (mm')^\varepsilon)$ (Part I, Corollary 2.7).

Part I deduces from (ii) that $r(n)r(N-n)$ has level of distribution $1/2$ (Theorem 3.2 there) and, with sieve parameters certified in interval arithmetic, that for every sufficiently large $N \equiv 2 \pmod 4$

$$\#\big\{1 \leq n < N :\ n = x^2+y^2,\ N-n = z^2+w^2,\ \Omega(n(N-n)) \leq 8,\ \max(\Omega(n), \Omega(N-n)) \leq 5\big\} \geq\ 0.23\, \mathfrak{S}_r(N)\, \frac{N}{(\log N)^2}, \qquad \mathfrak{S}_r(N) = \prod_{\substack{p \mid N \\ p > 2}} \frac{p-1}{p-2},$$

(Theorem 1.1 there; versions 1–2 had 9 and 6). The proof of (i) shows that the expansion of $\Theta_{m,m'}$ at every cusp is again a theta series of a form $A(x_1^2+x_2^2) + B(x_3^2+x_4^2)$ with $AB \asymp mm'$, whose coefficients are $O(n^\varepsilon)$ for $n \ll mm'$ because $A\alpha + B\beta = n$ has $O(1)$ solutions, and suggests $\|G_{m,m'}\|^2 \asymp (\log mm')^2$ (Part I, Remark 2.13); numerically, $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$ (Section 5). Theorem 1.5 below proves the lower bound, and Section 6 decomposes the norm over the cusps.

The first theme of this paper is the average over N of the error term, which is of independent interest and gives a stronger level of distribution for almost all N.

Theorem 1.2 (Mean square)**.** *Let $L \geq 1$, $G \in S_2(\Gamma_0(L))$ and $Y \geq 1$. Then*

$$\sum_{n \leq Y} |a_G(n)|^2 \ \leq\ 4\pi e^2\, Y \Big(1 + \frac{54\,Y}{L}\Big) \|G\|^2. \tag{1}$$

In particular, for coprime odd m, m', $G = G_{m,m'}$ and $Y \geq mm'$, by Theorem 1.1(i),

$$\sum_{N \leq Y} \big|T(m,m';N) - M(m,m';N)\big|^2 \ \ll_\varepsilon\ Y^2\, (mm')^{-1+\varepsilon}, \tag{2}$$

and hence, since $\sum_{Y/2 < N \leq Y} |a_G(N)|^2/N \leq (2/Y) \sum_{N \leq Y} |a_G(N)|^2$, the root mean square of $(T-M)/\sqrt{N}$ over $Y/2 < N \leq Y$ is $\ll (mm')^{-1/2+\varepsilon}$.

The proof of (1) is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor Y/L, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$. The bound (2) is the exact order of magnitude: by the Rankin–Selberg method the mean square of $(T-M)/\sqrt{N}$ is asymptotic to $16\pi^2\|G\|^2/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$, and Table 1 shows the data.

Averaging over N is natural for the arithmetic application, since the sieve of Part I is applied to one N at a time and an exceptional set of N can be tolerated. Let, as in Part I, $N\equiv 2 \pmod 4$, $\mathcal{A}=\{n\le N:\ (n,N)=1\}$, $w(n)=r(n)r(N-n)$, and for coprime square-free d,e with $(de,2N)=1$ let $R(d,e)=R(d,e;N)$ be the remainder in

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n,\ e\mid N-n}} w(n) = X_N\, h(d)h(e) + R(d,e;N)$$

(this is $S(d,e)=M(d,e)+R(d,e)$ of Part I, Section 3; we write X_N for the main term X there, to avoid a clash with the length X of the range of N below).

Theorem 1.3 (Level of distribution $1-\varepsilon$ for almost all N)**.** *Let $0<\varepsilon<1/20$, $X\ge X_0(\varepsilon)$ and $D=X^{1-3\varepsilon}$. Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}}\ \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\, d_3(de)\,|R(d,e;N)| \ \ll_\varepsilon\ X^{2-\varepsilon/2}.$$

Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ integers $N\equiv 2 \pmod 4$ in $(X/2,X]$,

$$\sum_{\substack{de\le N^{1-3\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\, d_3(de)\,|R(d,e;N)| \ \le\ N^{1-\varepsilon/5}. \tag{3}$$

The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at the level $\theta=19/20$ gives the following.

Corollary 1.4. *For all but $O(X^{1-\delta})$ integers $N\equiv 2 \pmod 4$ in $(X/2,X]$, with an absolute $\delta>0$,*

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 5,\ \max(\Omega(n),\Omega(N-n))\le 4\big\} \ge\ 0.34\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2},$$

the integers counted being coprime to N and products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{2/25}$.

The certified sieve parameters are $\alpha=2/25$, $\lambda=1/3$, $\beta_1=69/100$ (Section 3). The pairs $(5,4)$ for almost all N and $(8,5)$ for every N should be compared with $(9,6)$ for every N, which is what Waibel's bound alone gives (Part I, Remark 4.5). A floating-point scan (Figure 2) shows that the best attainable value of $1/\lambda+1/\beta_1$ drops below 4 only for $\theta>0.88$; at $\theta=19/20$ the scan gives 3.83 (at $\alpha=0.10$, $\beta_1=0.47$, $\lambda=0.59$, with $1/\lambda+2/\beta_1=5.96$), i.e. the pair $(5,3)$, which we have not certified, for the reason explained next.

Corollary 1.4 must also be compared with what the circle method gives for almost all N. By the method of Montgomery and Vaughan [11], carried out with the characters modulo 4 (see [4] for Goldbach's problem with primes in prescribed residue classes), all but $O(X^{1-\delta})$ integers $N\equiv 2 \pmod 4$ up to X are sums $p+q$ of two primes $p\equiv q\equiv 1 \pmod 4$, with $\gg \mathfrak{S}_r(N)N/(\log N)^2$ representations; such primes are sums of two squares, so for almost all N the pair $(1,1)$ holds with the count of Corollary 1.4. As an arithmetic statement about almost all N, Corollary 1.4 is therefore far from new. What is new is Theorem 1.3, a level of distribution $1-\varepsilon$ for the weighted

sequence $r(n)r(N-n)$, together with the observation that the sieve of Part I extracts $(5,4)$ from it with no further input; the sieve route adds something only for *every* N, where the circle method gives nothing (Goldbach's problem) and where Part I gives $(8,5)$. Section 4 formulates the individual bound that would raise the level to $1-\varepsilon$ for every N and with it give $(5,4)$ for every N (Proposition 4.2).

The third theme is the size of the norm itself. Theorem 1.1(i) is an upper bound. The proof in Part I shows that the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ is a theta series of a form $A(x_1^2+x_2^2)+B(x_3^2+x_4^2)$ with $AB \asymp mm'$; at the four cusps which are the images of ∞ under the Atkin–Lehner involutions W_m, W_{4m}, $W_{m'}$, $W_{4m'}$ this form is $x_1^2+x_2^2+mm'(x_3^2+x_4^2)$, whose first mm' representation numbers are the $r(n)$ themselves, and there $G_{m,m'}$ is transformed, exactly, into $G_{1,mm'}$. Integrating $|G|^2$ over the parts of the fundamental domain above these four cusps gives a lower bound.

Theorem 1.5 (Lower bound for the norm)**.** *Let m, m' be coprime odd positive integers with $mm' > 1$, and let $a(n) = a_{G_{1,mm'}}(n) = T(1,mm';n) - M(1,mm';n)$, so that $a(n) = r(n) - M(1,mm';n)$ for $1 \le n < mm'$.*

(i) $$\|G_{m,m'}\|^2 \;\ge\; \frac{1}{4\pi} \sum_{Q\in\{m,\,4m,\,m',\,4m'\}} \sum_{n\ge 1} \frac{|a(n)|^2}{n}\, e^{-4\pi n/Q}.$$

(ii) $\|G_{m,m'}\| = \|G_{1,mm'}\|$*: the norm depends only on the product mm'.*

(iii) *There is an absolute constant C such that*

$$\|G_{m,m'}\|^2 \;\ge\; \frac{1}{\pi}(\log mm')^2 - C\log(2mm')\,(\log\log 3mm')^C.$$

Part (ii) is the Atkin–Lehner involution W_m, which maps $\Theta_{m,m'}$ to $\chi_{-4}(m)\Theta_{1,mm'}$ and is an isometry; (iii) is (i) for the pair $(1,mm')$ at its two cusps of widths mm' and $4mm'$. Together with Theorem 1.1(i), $(\log mm')^2 \ll \|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for $mm' \ge C'$, and the exponent ε cannot be replaced by 0. In fact the factor $(mm')^\varepsilon$ can be removed altogether. Theorem 6.5 in Section 6.4 replaces the two ε-losses in the proof of Part I by mean values of $r(\alpha)^2$ and of the local densities and gives

$$\|G_{m,m'}\|^2 \ll (\log 3mm')^2(\log\log 3mm')^C,$$

$$\text{hence}\qquad \|G_{m,m'}\|^2 \asymp_k (\log mm')^2 \qquad \text{whenever } \omega(mm') \le k,$$

and Theorem 1.1(ii) holds with $(\log 3mm')(\log\log 3mm')^C$ in place of $(mm')^\varepsilon$.

On the spectral side, Theorem 6.7 makes the Rankin–Selberg structure of Part I, Section 5.5, explicit: for every newform φ of weight 2 and level $4mm'$ (mm' square-free),

$$\langle \Theta_{m,m'}, \varphi\rangle = \frac{4}{\pi^2}\,\chi_{-4}(m')\,\eta_{m'}(\varphi)\,L(1,\varphi)\,L(1,\varphi\otimes\chi_{-4}),$$

with no further local factors; so the norm bounds are statements about the second moment of $L(\tfrac12,\varphi)L(\tfrac12,\varphi\otimes\chi_{-4})$ over the family of level $4mm'$ (Corollary 6.10). The inequality (i) is nearly an equality when $m=1$: the four regions then cover the fraction $5/(2\pi) = 0.80$ of the area of $\Gamma_0(L)\backslash\mathbb{H}$, and numerically they carry 98% of the norm (Table 3). The proof is in Section 6, which also computes, for square-free m, m', the contribution of every cusp of the fundamental domain of the pair (m,m'): these cusp sums depend on the factorisation (for $m \approx m'$ they are $\frac{1}{\pi}((\log m)^2 + (\log m')^2) + \frac{5}{\pi^2}\log m \log m' + \cdots$, a sum over the divisors of mm') although their total, the norm, does not; the empirical law $0.37(\log mm')^2$ of Section 5 is the leading term $\frac{1}{\pi}(\log mm')^2$ of (iii) together with the positive lower-order terms, which are still large at $mm' \le 10^6$.

The fourth theme is the sieve itself. Part I sifts n and $N-n$ by the vector sieve of Brüdern and Fouvry, two linear sieves combined; but $n(N-n)$ is a sieve problem of dimension 2 (Remark 3.3), and the two-dimensional sieve of Diamond, Halberstam and Richert [8, 7], with Richert's weights, has the smaller sieving limit $\beta_2 = 4.2665$. Section 7 computes its sieve functions F_2, f_2 (we determine the constants $\alpha_2 = 5.3577$, $\beta_2 = 4.2665$ of [8] independently) and obtains, from the same level of distribution $1/2$ of Part I,

$$\Omega(n(N-n)) \le 7, \qquad \max(\Omega(n), \Omega(N-n)) \le 5$$

for $\ge 0.25\,\mathfrak{S}_r(N)N/(\log N)^2$ integers n, for every large $N \equiv 2 \pmod 4$ (Theorem 7.1): $(7,5)$ in place of $(8,5)$. The sieve functions enter through two numerical inequalities, $f_2(9) \ge 0.9995$ and a bound for an integral of F_2 (Lemma 7.2), which we certify in interval arithmetic from the constants α_2, β_2 computed in ball arithmetic by Booker and Browning [2], as Part I does for the linear sieve functions; the theorem is unconditional.

Section 4 restates the uniform bound of Theorem 1.1(ii) in terms of Poincaré series and formulates the individual bound, Section 5 contains the numerical evidence: the data of Part I, an exact check of the whole numerical chain at the level 20, the empirical law $\|G_{m,m'}\|^2 \approx 0.37(\log mm')^2$, which corrects a remark in the first two versions of Part I, and the supremum of $|T-M|/\sqrt{N}$ over N; Section 6 proves Theorem 1.5 and decomposes the norm over the cusps; Section 7 contains the two-dimensional sieve; and Section 8 describes the sum over the moduli that a level beyond $1/2$ for every N would require, in the form of a dispersion problem, with numerical evidence that its terms cancel like random signs.

2. Proof of Theorem 1.2

Let $G(z) = \sum_{n\ge1} a(n)e(nz) \in S_2(\Gamma_0(L))$ and put $F(z) = y^2|G(z)|^2$, $z = x+iy$. Since $G(\gamma z) = (cz+d)^2 G(z)$ and $\operatorname{Im}(\gamma z) = y/|cz+d|^2$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_0(L)$, the function F is $\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu = y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2 = \int_{\mathcal F} F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal F$ of $\Gamma_0(L)$.

Let $y_0 = 1/(4\pi Y)$ and $R = \{x+iy:\ 0 \le x < 1,\ y_0 \le y \le 2y_0\}$ (half-open in x, so that each orbit of the translations $z \mapsto z+b$ meets R at most once). By Parseval, $\int_0^1 |G(x+iy)|^2\,\mathrm{d}x = \sum_n |a(n)|^2 e^{-4\pi ny}$, hence

$$\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \sum_n |a(n)|^2 \int_{y_0}^{2y_0} e^{-4\pi ny}\,\mathrm{d}y \ \ge\ \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi ny \le 8\pi Y y_0 = 2$ for $n \le Y$, $y \le 2y_0$. On the other hand $\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu = \sum_{\gamma\in\Gamma_0(L)/\{\pm1\}} \int_{R\cap\gamma\mathcal F} F\,\mathrm{d}\mu = \int_{\mathcal F} F(w)\,\#\{\gamma:\ \gamma w \in R\}\,\mathrm{d}\mu(w) \ \le\ \mu_R\,\|G\|^2,$$

where $\mu_R = \sup_{z\in R}\#\{\gamma\in\Gamma_0(L)/\{\pm1\}:\ \gamma z\in R\}$ (if $\#\{\gamma:\gamma w\in R\}\ne0$ for some $w\in\mathcal F$, pick γ_0 with $z=\gamma_0 w\in R$; then $\gamma\mapsto\gamma\gamma_0^{-1}$ is a bijection from $\{\gamma:\gamma w\in R\}$ onto $\{\gamma:\gamma z\in R\}$, so the count at w is at most μ_R). It remains to bound μ_R. Fix $z = x+iy \in R$ and let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_0(L)$ with $\gamma z \in R$.

- If $c = 0$, then $\gamma = \pm\left(\begin{smallmatrix} 1 & b \\ 0 & 1\end{smallmatrix}\right)$ and $\gamma z = z+b$, so $x+b \in [0,1)$ forces $b = 0$: one element.
- If $c \ne 0$, we may take $c > 0$, since γ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L \mid c$, and $\operatorname{Im}(\gamma z) = y/|cz+d|^2 \in [y_0, 2y_0]$ together with $y \in [y_0, 2y_0]$ gives $|cz+d|^2 \in [1/2, 2]$. Since $|cz+d|^2 = (cx+d)^2 + c^2y^2$, this forces $c^2y^2 \le 2$, hence $c \le \sqrt{2}/y_0 = 4\sqrt{2}\,\pi Y$, and $|cx+d| \le \sqrt{2}$. The number of positive multiples c of L with $c \le 4\sqrt{2}\pi Y$ is $\lfloor 4\sqrt{2}\pi Y/L\rfloor \le 4\sqrt{2}\pi\, Y/L$; for each c there are at most 3 integers d with $|cx+d| \le \sqrt{2}$; and for each pair (c,d) the elements

of $\Gamma_0(L)$ with bottom row (c,d) are $\left(\begin{smallmatrix}1 & k\\ 0 & 1\end{smallmatrix}\right)\gamma_0$, $k\in\mathbb{Z}$, whose images $\gamma_0 z+k$ have real part in $[0,1)$ for exactly one k. This gives at most $4\sqrt{2}\pi\cdot 3\,Y/L = 12\sqrt{2}\pi\,Y/L < 54\,Y/L$ elements. Hence $\mu_R \le 1+54Y/L$, and

$$\sum_{n\le Y}|a(n)|^2 \le 4\pi e^2 Y\int_R F\,\mathrm{d}\mu \le 4\pi e^2\,Y\Big(1+\frac{54Y}{L}\Big)\|G\|^2,$$

which is (1). For $G=G_{m,m'}$, $L=4mm'$ and $Y\ge mm'$ the bracket is $\le 1+54Y/(4mm') \le 15Y/(mm')$, and Theorem 1.1(i) gives (2). □

Remark 2.1. The argument is the classical one for bounding $\sum_{n\le Y}|a(n)|^2$ by the norm (compare [14, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0\le y\le 2y_0$ with $y_0=1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at e^{-2} for all $n\le Y$. For $Y\ge L$ the factor Y^2/L in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y}|a(n)|^2/n \sim 16\pi^2\|G\|^2 Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$ with $\operatorname{vol}=\frac{\pi}{3}L\prod_{p\mid L}(1+1/p)$ (measure $\mathrm{d}\mu$), so that $\sum_{n\le Y}|a(n)|^2\sim 8\pi^2\|G\|^2Y^2/\operatorname{vol}$ for fixed G as $Y\to\infty$. Theorem 1.2 makes this uniform in the level, at the cost of the constant.

3. Proof of Theorem 1.3 and Corollary 1.4

3.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T-M|$ by Theorem 1.2 after an application of Cauchy–Schwarz over N. Recall from Part I, (16), that for $N\equiv 2 \pmod 4$ and coprime square-free d,e with $(de,2N)=1$,

$$R(d,e;N)=\sum_{t\mid N^\flat}\mu(t)\sum_{m\mid d^2}\sum_{m'\mid e^2}c_d(m)c_e(m')\sum_{\substack{t_1\mid t^2\\ t\mid t_1}}c_t(t_1)\big(a_G(N)-a_G(N/2)\big),\qquad G=G_{t_1 m,m'},\tag{4}$$

where $N^\flat$ is the product of the odd primes dividing N, $m=ds$, $m'=es'$ with $s\mid d$, $s'\mid e$, and likewise $t_1=tu$ with $u\mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (15), vanish unless $m=ds$ with $s\mid d$, resp. $t_1=tu$ with $u\mid t$; in particular $t\mid t_1$); the parameters t_1m,m' of the form $Q_{t_1m,m'}$ are coprime, odd and cube-free, $\tilde m\tilde m' := t_1mm' = t_1ss'\cdot de$, and the coefficients c satisfy $|c_d(m)c_e(m')c_t(t_1)|\le 3^{\omega(d)+\omega(e)+\omega(t)}$. (For the reader's convenience: c_d is the multiplicative coefficient of Part I, (15), with $c_p(p)=2$, $c_p(p^2)=-1$ for a split prime p and $c_q(q)=0$, $c_q(q^2)=1$ for an inert prime q; it encodes the identities $r(n)=2r(n/p)-r(n/p^2)$ and $r(n)=r(n/q^2)$, which turn the condition $d\mid n$ into a combination of theta series $\theta^2(mz)$. The sum over t is the Möbius inversion of the condition $(n,N)=1$, and $a_G(N)-a_G(N/2)$ selects the odd n.) Let $\eta=\varepsilon/10$. For $N\le X$ with $X\ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\,3^{\omega(d)+\omega(e)+\omega(t)}\,4^{\omega(N)}d(N)^2\le X^\eta$; for fixed (d,e,N) the number of tuples (s,s',t,t_1) is at most X^η; and $\sum_{k\le D}d(k)/k\le X^\eta$, $\sum_{k\le D}d(k)k^{-1/2}\le D^{1/2}X^\eta$. We also use the two bounds for a single term,

$$|a_G(N')|\le T(\tilde m,\tilde m';N')+M(\tilde m,\tilde m';N')\ll X^{2\eta}\Big(\frac{X}{\tilde m\tilde m'}+1\Big)\qquad (N'\le X),\tag{5}$$

$$\sum_{n\le X}|a_G(n)|\le X^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}\ll X\Big(1+\frac{X}{L}\Big)^{1/2}L^\eta,\tag{6}$$

where $G=G_{\tilde m,\tilde m'}$ and $L=4\tilde m\tilde m'$, the first being (21) of Part I, the second Cauchy–Schwarz, (1) and Theorem 1.1(i); for $L\le X$ the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/2+\eta}$. Let Σ denote the double sum in Theorem 1.3. Inserting (4) and splitting the tuples according to the size of t_1ss':

Tuples with $t_1ss' > X^\varepsilon$. For each N and each tuple, (5) with $\tilde m\tilde m' > X^\varepsilon de$ gives $|a_G(N')| \ll X^{2\eta}(X^{1-\varepsilon}/(de)+1)$. Summing over the $\le X^\eta$ tuples, over $N \le X$ and over $de \le D$ with the weights $d_3(de)\cdot 3^\omega \le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta}\sum_{k\le D} d(k)\Big(\frac{X^{1-\varepsilon}}{k}+1\Big) \ll X^{2-\varepsilon+5\eta} + X^{1+5\eta}D \ll X^{2-\varepsilon/2}.$$

Tuples with $t_1ss' \le X^\varepsilon$. Now $t \le t_1 \le X^\varepsilon$ (as $t \mid t_1$), so t runs over at most X^ε values, t_1 over at most $d(t^2) \le X^\eta$ values, and (s,s') over at most $d(d)d(e) \le X^\eta$ values; the forms $G = G_{t_1 m,m'}$ have level $L = 4t_1ss'de \le 4X^\varepsilon D \le X$, so that the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/2+\eta}$. For a fixed tuple and fixed (d,e), we drop the condition $t \mid N$ except for the count: the number of $N \le X$ with $t \mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\sum_{\substack{N\le X\\ t\mid N}} \big(|a_G(N)| + |a_G(N/2)|\big) \le 2\Big(\frac{X}{t}\Big)^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}$$
$$\ll \Big(\frac{X}{t}\Big)^{1/2} X\,(\tilde m\tilde m')^{-1/2+\eta} \le X^{3/2+\eta}\,t^{-1/2}(de)^{-1/2},$$

using (1) in the form $\sum_{n\le X}|a_G(n)|^2 \ll X^2\|G\|^2/L$ and $(\tilde m\tilde m')^{-1/2+\eta} \le X^\eta(de)^{-1/2}$. Summing over $t \le X^\varepsilon$ ($\sum_t t^{-1/2} \le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de \le D$ with the weights $\le X^\eta$,

$$\ll X^{3/2+4\eta+\varepsilon/2}\sum_{k\le D} d(k)k^{-1/2} \ll X^{3/2+5\eta+\varepsilon/2}D^{1/2} = X^{2-\varepsilon+5\eta} = X^{2-\varepsilon/2},$$

since $D^{1/2} = X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion. (The margin is small but positive: the choice $D = X^{1-3\varepsilon}$ saves $X^{-3\varepsilon/2}$ in $D^{1/2}$, of which the sum over t costs $X^{\varepsilon/2}$ and the divisor-type factors cost $X^{5\eta} = X^{\varepsilon/2}$; this is why $\eta = \varepsilon/10$.)

For the second, let $S(N)$ be the inner sum with $D = X^{1-3\varepsilon}$. The number of $N \in (X/2, X]$, $N \equiv 2 \pmod 4$, with $S(N) > X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4} \ll X^{1-\varepsilon/4}$. For the remaining N, since $N^{1-3\varepsilon} \le D$ and $X^{1-\varepsilon/4} \le (2N)^{1-\varepsilon/4} \le N^{1-\varepsilon/5}$ for large X, (3) holds. □

Remark 3.1. The exponent 1 is the balance $X^{3/2}D^{1/2} = X^2$ between the mean-square bound $X^{3/2}(de)^{-1/2}$ for the sum over N of one remainder and the number D of moduli; with a norm bound $\|G_{m,m'}\| \ll (mm')^{\beta+\varepsilon}$ in place of Theorem 1.1(i) one gets the level $1/(2(1+\beta))$ for every N (Part I, Remark 3.3) and $1/(1+2\beta)$ for almost all N; Waibel's $\beta = 1/4$ gives $2/5$ and $2/3$, and $\beta = 0$ gives $1/2$ and 1. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$; a bound for a higher moment would shrink it, and we do not pursue this.

3.2. **The sieve at the level 19/20.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with $2/5$); they hold verbatim, for a given N, with $\theta = 2/5$ replaced by any $\theta \in (0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{1-3\varepsilon}$, provided $\alpha < \beta_1 < \theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every N satisfying (3) with $3\varepsilon < 1/20$, the conclusion of Proposition 4.3 holds with $\theta = 19/20$, and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta = 19/20$. The script `certify_modular.py` of Part I, extended by the argument `19/20` in the repository of this paper, certifies in interval arithmetic

$$\theta = \tfrac{19}{20}, \qquad \alpha = \tfrac{2}{25}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{69}{100},$$
$$\tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{407}{69} < 6, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{307}{69} < 5, \qquad L - \lambda R \ge 0.085058$$

(with $L \geq 0.999484$, $R \leq 2.743278$, $\theta_A = 0.4740$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \leq 5$ and $\max(\Omega(n), \Omega(N-n)) \leq 4$, and the weight of each n counted is $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \leq 2^9$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.085058 + o(1)\big)\, 2^{-9} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{\alpha^2 (\log N)^2} \;\geq\; 0.34\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

since $0.085058 \cdot 2^{-9} \cdot 8 \cdot 1.66485 \cdot (25/2)^2 = 0.3457\ldots$. The exceptional set is that of Theorem 1.3 with, say, $\varepsilon = 1/80$, so $\delta = 1/320$ is admissible. □

Remark 3.2. The same sieve gives $(8,5)$ at the level $1/2$ (Part I), $(6,4)$ at $2/3$ (certified: $\alpha = 3/50$, $\lambda = 1/3$, $\beta_1 = 51/100$, $L - \lambda R \geq 0.036741$) and $(5,4)$ at $19/20$. The floating-point scan of Figure 2 shows that $1/\lambda + 2/\beta_1$ drops below 6 only for $\theta > 0.83$, and that the best attainable value of $1/\lambda + 1/\beta_1$ drops below 4 for $\theta > 0.88$: at $\theta = 19/20$ the minimum is 3.83, attained at $\alpha = 0.10$, $\beta_1 = 0.47$, $\lambda = 0.59$, where $1/\lambda + 2/\beta_1 = 5.96$, so that the Richert weights give $(5,3)$ for almost all N in the floating-point model, with the sifting parameter α at the edge of the scanned range. We have not certified this set, because Corollary 1.4 is in any case superseded for almost all N by the circle method (Section 1); it is an "almost all N" statement with a power-saving exceptional set and says nothing about any individual N, which is where the sieve route matters (Section 4.1).

Remark 3.3 (The vector sieve is not the natural sieve here). The sieve of Part I treats the two sides n and $N-n$ by the vector sieve of Brüdern and Fouvry, with two linear components of levels N^{θ_A} and $N^{\theta-\theta_A}$ and the lower bound $L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B)$. For $s_A, s_B \leq 3$ one has $f_1/F_1(s) = \log(s-1)$, so $L > 0$ if and only if $(s_A - 1)(s_B - 1) > e$, and with $s = s_A + s_B$ fixed this requires $s > 2 + 2\sqrt{e} = 5.297\ldots$, attained at the symmetric split. The product sequence $n(N-n)$, however, is a sieve problem of dimension 2 in the classical sense: under the weight $r(n)r(N-n)$ the density of $p \mid n(N-n)$ is $2h(p)$, $\sum_{p<z} 2h(p) = 2\log\log z + O(1)$, the condition $d \mid n(N-n)$ with $(n, N) = 1$ splits uniquely as $d = d_1 d_2$, $d_1 \mid n$, $d_2 \mid N-n$, and the remainder for the modulus d is $\sum_{d_1 d_2 = d} R(d_1, d_2)$, which is exactly what Theorem 3.2 of Part I and Theorem 1.3 bound (this is the reason for the weight $d_3(de)$ there). The two-dimensional sieve of Diamond, Halberstam and Richert [7] applies to it directly, with the joint level $d \leq N^{\theta}$ and the sieving limit $\beta_2 = 4.2665\ldots$ in place of 5.297; its upper and lower functions F_2, f_2 should be used in place of the products of F_1, f_1 both in L and in the weight term R. Section 7 carries this out: with the sieve functions F_2, f_2 enclosed in interval arithmetic, the two-dimensional sieve gives $(7,5)$ in place of $(8,5)$ for every N at the level $1/2$ (Theorem 7.1), $(6,4)$ at $2/3$ with a larger margin, and $(5,3)$ at $19/20$; the bound for $\max(\Omega(n), \Omega(N-n))$ comes from the same Richert weights, which see the two sides separately. For comparison, for the unweighted sequence $n(N-n)$, which has level of distribution 1 trivially, the two-dimensional weighted sieve gives $N = P_2 + P_3$ for every large even N (Wang Yuan [16]).

4. The uniform bound and the Poincaré series

Let $L = 4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N \geq 1$, be the Riesz representative of the N-th coefficient: the unique cusp form with

$$\langle f, P_N \rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{7}$$

i.e. $P_N = (4\pi N)^{-1} \sum_{f \in \mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the N-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define P_N by (7).) By

(11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2}\sum_{f\in\mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N,N)}{4\pi N},$$

where $\Delta_L(N,N)$ is the Petersson density. Hence, for $G = G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N\rangle|^2 = 4\pi N\, \Delta_L(N,N)\, \|G\|^2 \cos^2\angle(G,P_N), \tag{8}$$

where $\cos\angle(G,P_N) = |\langle G,P_N\rangle|/(\|G\|\,\|P_N\|)$. The identity separates the three quantities that enter any bound for $a_G(N)$: the norm $\|G\|^2$, which by Theorem 1.1(i) is $\ll L^{\varepsilon}$; the Petersson density, which is $\le C'$ for $N \le L^{1/2}$ (Part I, Lemma 2.2, Step 4) and $1 + O(N^{1/2+\varepsilon}/L)$ in general; and the alignment $\cos^2\angle(G,P_N) \le 1$. Theorem 1.1(ii) is obtained from (8) by bounding the alignment trivially: after the coprime part of N is moved into a Hecke operator (Part I, Lemma 2.2), only P_{n_2} with $n_2 \in \{1,2\}$ occurs, where the density is bounded, and the norm does the rest. Theorem 1.2 says what the alignment is on average: inserting (8) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2}\sum_{Y/2<N\le Y} \Delta_L(N,N)\cos^2\angle(G,P_N) \;\le\; 4e^2\Big(\frac{1}{Y}+\frac{54}{L}\Big) \;\ll\; \frac{1}{L} \qquad (Y \ge L), \tag{9}$$

the value of $\cos^2$ for a unit vector in general position in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$: on average over N, the theta series is no more aligned with the Poincaré series than a random cusp form of level L would be, and Theorem 1.1(ii) says that no single N is aligned by more than $N^{\varepsilon}L/\|G\|^2 \gg L^{1-\varepsilon}$ times the average. Expanding in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N\rangle = \sum_{\varphi\in\mathcal{B}^*} \frac{\langle G,\varphi\rangle\langle\varphi,P_N\rangle}{\|\varphi\|^2} = \frac{1}{4\pi N}\sum_{\varphi\in\mathcal{B}^*} \frac{\langle\Theta,\varphi\rangle\, a_\varphi(N)}{\|\varphi\|^2},$$

so that Theorem 1.1(ii) is the bound $\ll N^{1/2+\varepsilon}$ for this spectral sum, uniformly in m, m'; the first two versions of Part I asked for a proof through the cancellation between the coefficients $\langle\Theta,\varphi\rangle$ and $a_\varphi(N)$. The proof in Part I does not see this cancellation: it shows instead that $\sum_\varphi |\langle\Theta,\varphi\rangle|^2/\|\varphi\|^2 = \|G\|^2$ is itself small, i.e. that the theta series has a small cuspidal projection to begin with, and the spectral sum is then bounded by Cauchy–Schwarz.

4.1. **The individual bound.** Theorem 1.2 and the data (Section 5.4) say that the cancellation which the proof does not see is there for most N, and the sieve for every N would profit from it for each N. We state the strongest form suggested by the data.

Conjecture 4.1. For coprime odd m, m' and $N \ge 1$ with $(N, mm') = 1$ and $v_2(N) \le 1$,

$$|T(m,m';N) - M(m,m';N)| = |a_{G_{m,m'}}(N)| \;\ll_\varepsilon\; N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}.$$

In the terms of (8): $\Delta_L(N,N)\cos^2\angle(G,P_N) \ll N^{\varepsilon}L^{-1+\varepsilon}/\|G\|^2$, i.e. no Poincaré series P_N is aligned with G by more than $N^{\varepsilon}L^{\varepsilon}$ times the average (9).

Conjecture 4.1 is the mean-square bound (2) applied to each N separately; it is consistent with the trivial lower bound $|a_G(N)| \ge \operatorname{dist}(M, 8\mathbb{Z})$ for $N \ll L$, where $M \asymp N/(mm') \ll 1$, and with the data of Section 5.4, where the supremum over 10^6 to $4\cdot 10^6$ values of N exceeds the root mean square by a factor between 5 and 10 at every level computed. We do not know how to prove any power saving in mm' for an individual N. In the spectral form the question is a twisted first moment $\sum_\varphi \langle\Theta,\varphi\rangle a_\varphi(N)/\|\varphi\|^2$ over the forms of level L with the "long" twist $N \ge L$ (the sieve at the level θ uses $N \approx L^{1/\theta}$), outside the range $N \le L^{1-\delta}$ in which the Petersson formula evaluates such moments. Any saving would raise the level for every N:

Proposition 4.2. *Let $0 \le \eta \le \frac{1}{2}$ and suppose that $|a_{G_{m,m'}}(N)| \ll_\varepsilon N^{1/2+\varepsilon}(mm')^{-\eta+\varepsilon}$ for all coprime odd m, m' and all N with $(N, mm') = 1$, $v_2(N) \le 1$. Then Theorem 3.2 of Part I holds for every $N \equiv 2 \pmod 4$ with $D = N^{\theta-\varepsilon}$, $\theta = 1/(2(1-\eta))$, i.e. $r(n)r(N-n)$ has level of distribution $1/(2(1-\eta))$. In particular, Conjecture 4.1 gives the level $1-\varepsilon$ for every N, and then the sieve of Section 3 gives, for every sufficiently large $N \equiv 2 \pmod 4$, $\Omega(n(N-n)) \le 5$ and $\max(\Omega(n), \Omega(N-n)) \le 4$ for at least $0.34\,\mathfrak{S}_r(N)N/(\log N)^2$ integers n.*

Proof. We follow the proof of Theorem 3.2 of Part I and only change the bound for one remainder. Each term of $R(d,e)$ is $\pm c\, a_G(N')$ for a form $G = G_{\tilde m,\tilde m'}$ with $\tilde m\tilde m' = t_1 ss'\cdot de$, where $t \mid N^\flat$, $t_1 \mid t^2$, and $N' \in \{N, N/2\}$ may have common factors with $\tilde m\tilde m'$ at the primes of t only. Lemma 2.5 of Part I reduces $a_G(N')$ to $\le 4^{\omega(N)}d(N)$ terms $a_{G_{m_j,m_j'}}(N_j)$ with $(N_j, m_jm_j') = 1$, $N_j \mid N'$ and $m_jm_j' \ge \tilde m\tilde m'/t_1^2 \ge de/t_1^2$ (each reduction step at a prime p divides the product of the coefficients by at most p^2, and only the primes of t_1 are involved). Hence, by the hypothesis with $\varepsilon/20$ in place of ε,

$$|a_G(N')| \ll 4^{\omega(N)}d(N)\, N^{1/2+\varepsilon/20}\,(de)^{-\eta}\, t_1^{2\eta}\,(mm')^{\varepsilon/20}.$$

Put $\eta' = \varepsilon/20$. For the terms with $t_1ss' \le N^\varepsilon$ this is $\ll N^{1/2+3\eta'+2\varepsilon\eta}(de)^{-\eta}$ by (F1) of Part I, and their total is

$$\ll N^{1/2+5\eta'+\varepsilon}\sum_{k\le D} d(k)k^{-\eta} \ll N^{1/2+6\eta'+\varepsilon}D^{1-\eta} = N^{1/2+6\eta'+\varepsilon}N^{1/2-\varepsilon(1-\eta)} \le N^{1-\varepsilon/2+6\eta'},$$

since $D^{1-\eta} = N^{(\theta-\varepsilon)(1-\eta)} = N^{1/2-\varepsilon(1-\eta)}$ and $1-\eta \ge \frac12$. The terms with $t_1ss' > N^\varepsilon$ are bounded by the trivial bound (5) exactly as in Part I, by $N^{1-\varepsilon+5\eta'} + N^{\theta+5\eta'}$. With $\theta \le 1-\varepsilon$ (replace ε by $\min(\varepsilon, 1-\theta)$ if necessary) the total is $\ll N^{1-\varepsilon/2+6\eta'} \le N^{1-\varepsilon/5}$, which is the conclusion of Theorem 3.2 of Part I with $N^{1-\varepsilon/5}$ in place of $N^{1-\varepsilon/3}$; Proposition 4.3 of Part I needs only a power saving. The last assertion is Section 3, where the parameters at the level 19/20 were certified for any N satisfying (3). □

The exponent $\theta = 1/(2(1-\eta))$ interpolates between the level 1/2 of Part I ($\eta = 0$) and the level 1 ($\eta = \frac12$); the level 2/3, which gives (6, 4) for every N with the certified parameters of Section 3, needs only $\eta = \frac14$, i.e. a saving of $(mm')^{-1/4}$ over Theorem 1.1(ii), half of what the mean square (2) gives on average.

5. Numerical evidence

5.1. **The mean square and the norm.** The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y) = \sum_{N\in\mathcal{N}_Y}|a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{N \le Y :\ (N, mm') = 1,\ v_2(N) \le 3\},$$

on which the main term of that script is exact, for $Y = 2^{21}$ and $Y = 2^{20}$, and the estimate

$$\|G\|^2_{\mathrm{est}} = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2}\cdot\frac{1}{\#\mathcal{N}_Y}\sum_{N\in\mathcal{N}_Y}\frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 2.1. Replacing $1/Y$ by $1/\#\mathcal{N}_Y$ assumes that the mean square of $|a_G(N)|^2/N$ over the excluded N (those with $(N, mm') > 1$ or $v_2(N) \ge 4$) is the same as over $\mathcal{N}_Y$; this is immaterial when mm' has only large prime factors, but introduces a bias of unknown sign for $(1,5)$, $(9,1)$, $(3,5)$ and $(1,15015)$, where 25%, 38%, 50% and 64% of the N are excluded. Table 1 shows the result at $Y = 2^{21}$, and Figure 1 the dependence on Y and on mm'.

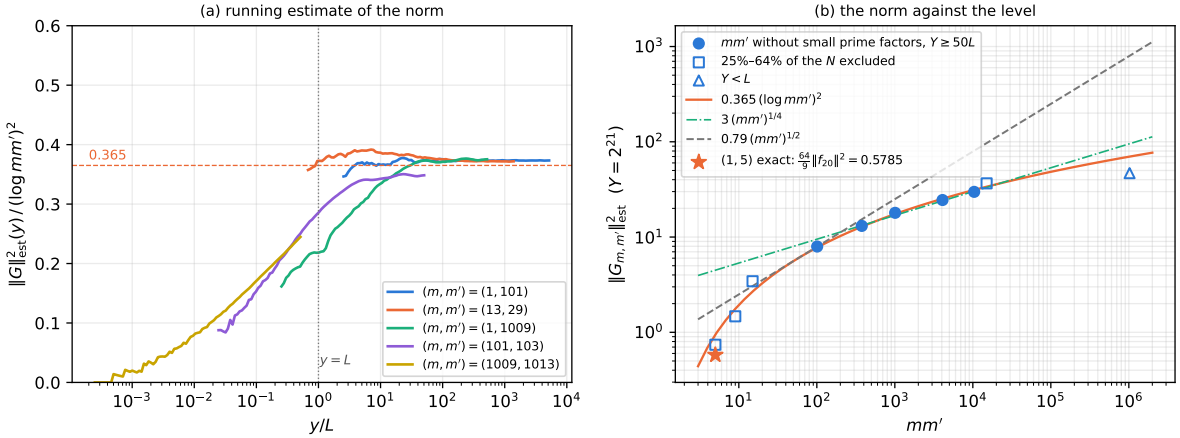


FIGURE 1. (a) The running estimate $\|G\|^2_{\mathrm{est}}(y)$, computed over the admissible $N \le y$, divided by $(\log mm')^2$, against y/L on a logarithmic scale, for five pairs (m, m'); the plateau for $y \gg L$ is the growth Y^2 of Theorem 1.2 and the Rankin–Selberg asymptotic of Remark 2.1, and its height is the constant 0.365 of Section 5.3 (dashed). (b) $\|G_{m,m'}\|^2_{\mathrm{est}}$ at $Y = 2^{21}$ against mm', both axes logarithmic; filled circles: the five pairs with mm' free of small prime factors and $Y \ge 50L$; open squares: pairs with 25%–64% of the N excluded; triangle: $(1009, 1013)$, for which $Y < L$; star: the exact value at $(1, 5)$. The lines are $0.365(\log mm')^2$, the fit of Section 5.3, $3(mm')^{1/4}$, and $0.79(mm')^{1/2}$ (the growth allowed by Waibel's general bound, through the point $(1, 101)$). Script `plot_mean_square.py`; floating point.

(m, m')	mm'	$\#\mathcal{N}_Y$	$S_2(Y)/S_2(Y/2)$	$\|G\|^2_{\mathrm{est}}$	$\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$	$\|G\|^2_{\mathrm{est}}/(\log mm')^2$
$(1, 5)$	5	1572864	4.001	0.74	0.33	0.29
$(9, 1)$	9	1310720	4.001	1.47	0.49	0.30
$(3, 5)$	15	1048576	4.001	3.44	0.89	0.47
$(1, 101)$	101	1946614	4.000	7.95	0.79	0.373
$(13, 29)$	377	1752262	4.001	13.09	0.67	0.372
$(1, 1009)$	1009	1964131	3.976	17.96	0.57	0.375
$(61, 67)$	4087	1904986	3.961	24.47	0.38	0.354
$(101, 103)$	10403	1927715	3.948	29.81	0.29	0.348
$(1, 15015)$	15015	754220	4.359	36.62	0.30	0.396
$(1009, 1013)$	1022117	1962192	4.559	46.87	0.046	0.244

TABLE 1. Mean square of the cuspidal error over the admissible $N \le Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth Y^2 of Theorem 1.2. For the first three pairs and for $(1, 15015)$ a large proportion of the N (25% to 64%) is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its Y^2/L form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness. The exact value for $(1, 5)$ is 0.5785 (Section 5.2).

5.2. **An exact check at the level 20.** The space $S_2(\Gamma_0(20))$ is one-dimensional, spanned by the newform $f_{20} = \eta(2z)^2\eta(10z)^2$ of the elliptic curve $20a1$: $y^2 = x^3 + x^2 + 4x + 4$, which is $X_0(20)$

itself (modular degree 1, Manin constant 1). Hence $G_{1,5} = c\, f_{20}$ with $c = a_G(1) = T(1,5;1) - M(1,5;1) = 4 - \frac{4}{3} = \frac{8}{3}$ (the Siegel product at $n=1$ is $\frac{\pi^2}{5} \cdot \frac{25}{3\pi^2} \cdot \frac{4}{5}$), and $\|G_{1,5}\|^2 = \frac{64}{9}\|f_{20}\|^2$. The script `norm_level20.py` computes $\|f_{20}\|^2$ in two independent ways: from the Rankin–Selberg asymptotic $\sum_{n\le y} a_f(n)^2/n \sim 16\pi^2\|f\|^2 y/\operatorname{vol}$, $\operatorname{vol}(\Gamma_0(20)\backslash\mathbb{H}) = 12\pi$, with the coefficients of the eta product up to 2^{20}, which gives 0.08128, 0.08128, 0.08134 at $y = 2^{16}, 2^{18}, 2^{20}$; and from the period lattice Λ of $20a1$, since $\varphi^*\omega = 2\pi i f(z)\mathrm{d}z$ gives $\|f\|^2 = \operatorname{covol}(\Lambda)/(4\pi^2)$ for a modular parametrisation of degree 1. The periods $\omega_1 = 2.824375\ldots$ (real) and $\omega_2 = 2.274165\ldots i$ of $y^2 = (x+1)(x^2+4)$ are computed by quadrature, the lattice $\mathbb{Z}\omega_1 + \mathbb{Z}(\omega_1+\omega_2)/2$ is identified as the period lattice by recomputing $g_2 = -44/3$ and $|g_3| = 296/27$ from it through E_4 and E_6, and its covolume $3.2115478\ldots$ gives $\|f_{20}\|^2 = 0.0813495\ldots$. The two values agree to four decimals. The script also checks that $T(1,5;N) - M(1,5;N)$ as computed by the pipeline of Part I equals $\frac{8}{3}a_f(N)$ on every admissible $N \le 2^{20}$, to 10^{-9}. This validates the Siegel main term, the FFT computation, the Rankin–Selberg normalisation and the volume used in the estimate $\|G\|^2_{\mathrm{est}}$. The exact value $\|G_{1,5}\|^2 = 0.5785$ is below the estimate 0.74 of Table 1, as expected: the excluded N are either multiples of 5, where $|a_f(5n)| = |a_f(n)|$, or divisible by 16, where $a_f = 0$, so the admissible mean is biased upwards. With the main term made exact for every N (Section 6.6), the estimate over all $N \le 2^{21}$ is 0.5785, in agreement with the exact value to four decimals.

5.3. **The empirical law.** The last column of Table 1 shows $\|G\|^2_{\mathrm{est}}/(\log mm')^2$. For the five pairs with $10^2 \le mm' \le 1.1\cdot 10^4$, $Y \ge 50L$ and mm' free of small prime factors, it lies between 0.348 and 0.375, with mean 0.365, while the ratio to $(mm')^{1/2}$ falls from 0.79 to 0.29 and the ratio to $(mm')^{1/4}$ rises from 2.5 to 3.2 over the same range. Over a range of a factor 100 in mm' the fits $(\log mm')^2$ and $(mm')^{1/4}$ are not separated by the data alone, but Theorem 1.1(i) excludes any power, and the heuristic of Part I, Remark 2.13, that $\|G\|^2$ is a sum over the cusps of $\sum_{n\ll mm'} |b(n)|^2/n$ with $|b(n)| \approx r(\alpha)r(\beta)$ on $O(1)$ pairs, predicts $(\log mm')^2$ with a constant of the order of $\frac{1}{4\pi}\cdot 2 \cdot$ (number of cusp classes with a large contribution) ≈ 0.3–0.5. We therefore record

$$\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2 \qquad (10^2 \le mm' \le 1.6\cdot 10^4),$$

and, for the root mean square of $(T-M)/\sqrt{N}$ over $N \le Y$, $Y \gg L$,

$$\mathrm{rms}^2 \approx \frac{16\pi^2\|G\|^2}{\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})} \approx \frac{44\,(\log mm')^2}{mm'\prod_{p\mid 2mm'}(1+1/p)},$$

which reproduces the column "rms" of Table 3 of Part I (0.27 for $(101,103)$, 0.67 for $(1,1009)$). This corrects the remark in Section 5.3 of the first two versions of Part I that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": the decrease is like $(mm')^{-1/2}$ up to a logarithm, as Theorem 1.2 with Theorem 1.1(i) shows; version 3 of Part I states the corrected law. Section 6 shows what the "law" is: the norm depends only on mm' (Theorem 1.5(ii)), its leading term is $\frac{1}{\pi}(\log mm')^2$, and the constant 0.37 is this leading term together with lower-order terms that are still large in the range of the sample; the estimates of Table 1 are themselves biased by the restriction to the admissible N (Table 3 has the unbiased values).

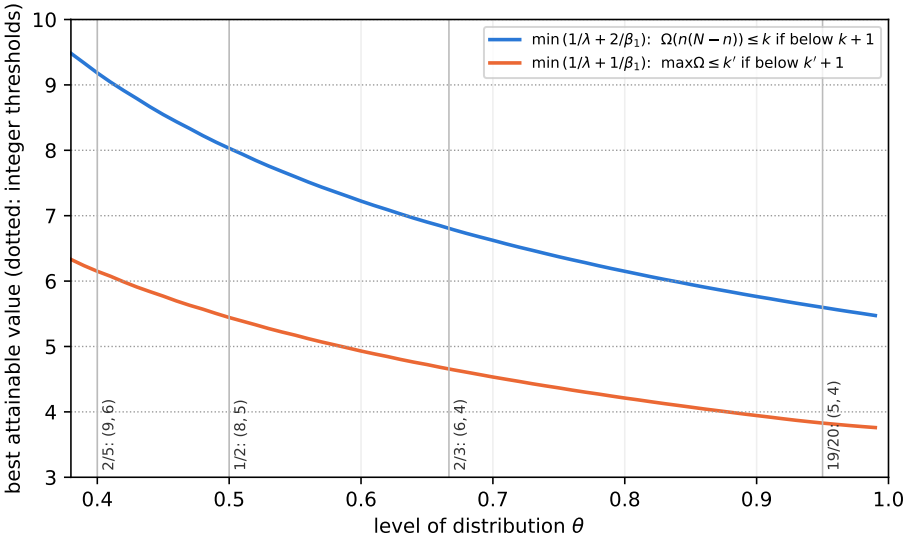


FIGURE 2. The smallest values of $1/\lambda + 2/\beta_1$ and of $1/\lambda + 1/\beta_1$ attainable with $L - \lambda R > 0$ in the sieve of Part I, as functions of the level of distribution θ (floating-point model of `modular_sieve_scan.py`, minimised over α, β_1 with $\lambda = L/R$; script `plot_sieve_thresholds.py`). The bound $\Omega(n(N-n)) \le k$, resp. $\max \Omega \le k'$, holds when the blue, resp. orange, curve is below the dotted line $k+1$, resp. $k'+1$. The vertical lines mark $\theta = 2/5$ (Waibel's bound alone) and $1/2$ (Part I, every N), $2/3$ and $19/20$ (almost all N); the parameter sets at these four levels are certified.

5.4. **The supremum over N.** Conjecture 4.1 says that the supremum of $|a_G(N)|/\sqrt{N}$ over N in a range is, up to $N^{\varepsilon}L^{\varepsilon}$, of the size of the root mean square $\approx 2.7(\log mm')(mm')^{-1/2}$ of Section 5.3. Table 2, extracted from the scans of Part I (`results/sup_error_scan_x21.txt`, `_x23.txt` there), gives the ratio of the supremum over all admissible $N \in (X/2, X]$ to the root mean square, for $X = 2^{21}$ (about 10^6 values of N per pair) and $X = 2^{23}$ (about $4 \cdot 10^6$). For the pairs with $L \le 6 \cdot 10^4$ the ratio lies between 5.4 and 9.8 and shows no trend in mm'; its mild growth from $X = 2^{21}$ to 2^{23} is the growth of the extreme value of a fixed heavy-tailed distribution with the sample size (the suprema are attained at N with 16–192 divisors, in line with the factor $d(N)$ of Deligne's bound). The pair $(9, 1)$ is special: $S_2(\Gamma_0(36))$ is spanned by the newform of the curve $36a1$, $y^2 = x^3 + 1$, which has complex multiplication, so that $a_f(p) = 0$ for $p \equiv 2 \pmod 3$ and the distribution of $a_f(N)$ is far from the Gaussian bulk of Figure 3 of Part I. The pair $(1009, 1013)$ is in the range $X < 2L$, where neither the mean square nor the supremum has reached its asymptotic regime.

(m,m')	mm'	rms	sup, $X=2^{21}$	sup /rms	sup, $X=2^{23}$	sup /rms
$(1,5)$	5	1.76	14.7	8.4	17.3	9.8
$(3,5)$	15	1.90	10.5	5.5	12.2	6.4
$(1,101)$	101	1.40	11.8	8.4	12.1	8.6
$(13,29)$	377	0.89	5.9	6.6	6.8	7.7
$(1,1009)$	1009	0.67	4.4	6.5	5.3	8.0
$(61,67)$	4087	0.38	2.5	6.5	2.4	6.3
$(101,103)$	10403	0.27	1.9	7.3	1.8	6.7
$(1,15015)$	15015	0.18	1.0	5.7	–	–
$(9,1)$	9	1.76	28.7	16.3	33.8	19.2
$(1009,1013)$	1022117	0.036	0.61	17	0.55	14

TABLE 2. The supremum of $|T(m,m';N)-M(m,m';N)|/\sqrt{N}$ over the admissible $N\in(X/2,X]$ of Part I, Section 5, and its ratio to the root mean square over the same N (the rms is the same to two decimals for both X). The last two pairs are the degenerate cases discussed in the text.

6. A lower bound for the norm: the regions above the cusps

Throughout this section m,m' are coprime odd positive integers, $L=4mm'$, $\chi=\chi_{-4}$, and for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in \mathrm{GL}_2^+(\mathbb{Q})$ and $k\in\{1,2\}$ we write $(f|_k\gamma)(z)=(\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z)$; thus scalar matrices act trivially, $f(mz)=m^{-k/2}(f|_kB_m)(z)$ with $B_m=\left(\begin{smallmatrix}m&0\\0&1\end{smallmatrix}\right)$, and $(fg)|_2\gamma=(f|_1\gamma)(g|_1\gamma)$. We use $\theta^2\in M_1(\Gamma_0(4),\chi)$, i.e. $\theta^2|_1\gamma=\chi(d)\theta^2$ for $\gamma\in\Gamma_0(4)$, and $\theta(-1/(4z))=\sqrt{-2iz}\,\theta(z)$, which gives $\theta^2|_1\left(\begin{smallmatrix}0&-1\\4&0\end{smallmatrix}\right)=-i\theta^2$.

6.1. **The Atkin–Lehner cusps.** For $Q\mid L$ with $(Q,L/Q)=1$, an Atkin–Lehner matrix is $W_Q=\left(\begin{smallmatrix}Qa&b\\Lc&Qd\end{smallmatrix}\right)$ with $a,b,c,d\in\mathbb{Z}$ and $\det W_Q=Q$ [1]. It normalises $\Gamma_0(L)$; hence for $f\in M_2(\Gamma_0(L))$ the form $f|_2W_Q$ lies in $M_2(\Gamma_0(L))$ and does not depend on the choice of W_Q (two choices differ by an element of $\Gamma_0(L)$ on either side), $f\mapsto f|_2W_Q$ is an isometry for the Petersson product which maps cusp forms to cusp forms, and it therefore preserves the orthogonal decomposition of $M_2(\Gamma_0(L))$ into the Eisenstein and the cuspidal subspace.

Lemma 6.1. *Let $Q\in\{m,4m\}$. Then $\Theta_{m,m'}|_2W_Q=\varepsilon_Q\,\Theta_{1,mm'}$ with $|\varepsilon_Q|=1$, and consequently $E_{m,m'}|_2W_Q=\varepsilon_QE_{1,mm'}$ and $G_{m,m'}|_2W_Q=\varepsilon_QG_{1,mm'}$. The same holds with m and m' interchanged.*

Proof. Let first $Q=m$ and $W_m=\left(\begin{smallmatrix}ma&b\\Lc&md\end{smallmatrix}\right)$, $mad-4m'bc=1$. Then $B_mW_m=m\left(\begin{smallmatrix}ma&b\\4m'c&d\end{smallmatrix}\right)$ with the second factor in $\Gamma_0(4)$, so $\theta^2(m\cdot)|_1W_m=m^{-1/2}\theta^2|_1(B_mW_m)=m^{-1/2}\chi(d)\theta^2$; and $W_m=\left(\begin{smallmatrix}a&b\\4m'c&md\end{smallmatrix}\right)B_m$ with the first factor in $\Gamma_0(4m')$, so $\theta^2(m'\cdot)|_1W_m=\chi(md)\,\theta^2(m'\cdot)|_1B_m=\chi(md)\,m^{1/2}\theta^2(mm'\cdot)$. Multiplying, $\Theta_{m,m'}|_2W_m=\chi(m)\,\theta^2(z)\theta^2(mm'z)=\chi(m)\Theta_{1,mm'}$.

Let now $Q=4m$ and $W_{4m}=\left(\begin{smallmatrix}4ma&b\\Lc&4md\end{smallmatrix}\right)$, $4mad-m'bc=1$. Then $B_mW_{4m}=m\left(\begin{smallmatrix}4ma&b\\4m'c&4d\end{smallmatrix}\right)$, and the second factor is an Atkin–Lehner matrix W_4 for $\Gamma_0(4)$; every such matrix is $\left(\begin{smallmatrix}0&-1\\4&0\end{smallmatrix}\right)$ times an element of $\Gamma_0(4)$, so $\theta^2|_1W_4=\pm i\theta^2$ and $\theta^2(m\cdot)|_1W_{4m}=\pm i\,m^{-1/2}\theta^2$. Also $W_{4m}=W_4'B_m$ with $W_4'=\left(\begin{smallmatrix}4a&b\\4m'c&4md\end{smallmatrix}\right)$ an Atkin–Lehner matrix for $\Gamma_0(4m')$, and $B_{m'}W_4'=\left(\begin{smallmatrix}4a&m'b\\4c&4md\end{smallmatrix}\right)B_{m'}$ with an Atkin–Lehner matrix for $\Gamma_0(4)$ in front, so $\theta^2(m'\cdot)|_1W_4'=m'^{-1/2}(\pm i\theta^2)|_1B_{m'}=\pm i\,\theta^2(m'\cdot)$ and $\theta^2(m'\cdot)|_1W_{4m}=\pm i\,m^{1/2}\theta^2(mm'\cdot)$. Multiplying, $\Theta_{m,m'}|_2W_{4m}=\pm\Theta_{1,mm'}$. The assertions on E and G follow because $|_2W_Q$ preserves the orthogonal decomposition and $\Theta_{1,mm'}=E_{1,mm'}+G_{1,mm'}$ is that decomposition for $\Theta_{1,mm'}$ (Siegel's theorem applied to $Q_{1,mm'}$). □

We now use the formula of Part I, Lemma 2.8 (Waibel's [17, (22)]): with $\mu=[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(L)]$, $\tilde\mu=[\mathrm{SL}_2(\mathbb{Z}):\Gamma(L)]$, $\tilde\mu/\mu=L\varphi(L)$, $\mathcal{F}$ the standard fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$, $T=\{\left(\begin{smallmatrix}1&t\\0&1\end{smallmatrix}\right):$

$0 \le t \le L-1\}$ and $\varrho_{a,c} = \left(\begin{smallmatrix} a & * \\ c & * \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$,

$$\|G\|^2 = \frac{\mu}{\tilde\mu} \sum_{c \ (L)} \sum_{\substack{a \ (L) \\ (a,c,L)=1}} I(\varrho_{a,c}), \qquad I(\varrho) = \int_{\bigcup_{t\in T} t\mathcal{F}} \bigl|(G|_2\varrho)(z)\bigr|^2 \,\mathrm{d}x\,\mathrm{d}y, \tag{10}$$

for $G \in S_2(\Gamma_0(L))$, where (a,c) runs over coprime representatives of the residue classes. Every term is non-negative, and $\bigcup_{t\in T} t\mathcal{F} \supseteq \{-\frac12 \le x \le L-\frac12,\ y \ge 1\}$.

Lemma 6.2. *Let $Q \in \{m, 4m, m', 4m'\}$ and $a(n) = a_{G_{1,mm'}}(n)$. The terms of (10) for $G = G_{m,m'}$ with $(c,L) = L/Q$ contribute in total at least*

$$\frac{1}{4\pi} \sum_{n\ge1} \frac{|a(n)|^2}{n}\, e^{-4\pi n/Q}.$$

Proof. Let $(c,L) = L/Q$, $c = (L/Q)c_1$ with $(c_1, Q) = 1$, and $(a,c) = 1$; then $(a, L/Q) = 1$, so $(aQ, (L/Q)c_1) = 1$ and there are $b, d \in \mathbb{Z}$ with $aQd - b(L/Q)c_1 = 1$. The matrix $\varrho = \left(\begin{smallmatrix} a & b \\ (L/Q)c_1 & Qd \end{smallmatrix}\right)$ is in $\mathrm{SL}_2(\mathbb{Z})$, has the prescribed first column, and $\varrho B_Q = \left(\begin{smallmatrix} Qa & b \\ Lc_1 & Qd \end{smallmatrix}\right)$ is an Atkin–Lehner matrix W_Q. Hence, by Lemma 6.1 and $B_Q^{-1} = Q^{-1}\left(\begin{smallmatrix} 1 & 0 \\ 0 & Q \end{smallmatrix}\right)$,

$$G_{m,m'}|_2\varrho = (G_{m,m'}|_2 W_Q)|_2 B_Q^{-1} = \varepsilon_Q\, G_{1,mm'}|_2 \left(\begin{smallmatrix} 1 & 0 \\ 0 & Q \end{smallmatrix}\right) = \frac{\varepsilon_Q}{Q}\, G_{1,mm'}\Bigl(\frac{z}{Q}\Bigr) = \frac{\varepsilon_Q}{Q} \sum_{n\ge1} a(n)\, e\Bigl(\frac{nz}{Q}\Bigr).$$

This function has period Q in x, and $Q \mid L$, so by Parseval over $-\frac12 \le x \le L-\frac12$,

$$I(\varrho) \ \ge\ \int_1^\infty \int_{-1/2}^{L-1/2} |G|_2\varrho|^2 \,\mathrm{d}x\,\mathrm{d}y = \frac{L}{Q^2} \sum_{n\ge1} |a(n)|^2 \int_1^\infty e^{-4\pi n y/Q}\,\mathrm{d}y = \frac{L}{4\pi Q} \sum_{n\ge1} \frac{|a(n)|^2}{n} e^{-4\pi n/Q}.$$

The number of pairs (a,c) modulo L with $(c,L) = L/Q$ and $(a,c,L) = 1$, i.e. $(a, L/Q) = 1$, is $\varphi(Q)\cdot\varphi(L/Q)\,Q = \varphi(L)Q$, and $\mu/\tilde\mu = 1/(L\varphi(L))$. Multiplying gives the assertion. □

Proof of Theorem 1.5(i) and (ii). Part (ii): by Lemma 6.1 applied to each prime of m in turn (or directly, with W_m of level L), $G_{m,m'}|_2 W_m = \varepsilon G_{1,mm'}$ with $|\varepsilon| = 1$, and $|_2 W_m$ is an isometry of $S_2(\Gamma_0(L))$. For (i): since m, m' are odd and coprime with $mm' > 1$, we have $m \ne m'$, $m \ne 4m'$ and $m' \ne 4m$, so the four values L/Q, $Q \in \{m, 4m, m', 4m'\}$, are distinct and the four sets of pairs (a,c) in Lemma 6.2 are disjoint. (When $m = 1$ the values $Q = 1$ and $Q = 4$ occur: $W_1 \in \Gamma_0(L)$, the class $(c,L) = L$ is the cusp ∞ itself and the expansion is that of $G_{1,m'}$; Lemmas 6.1 and 6.2 cover these cases as stated, since they use only the Atkin–Lehner matrices and Parseval's identity, and the four cusps are then all the Atkin–Lehner cusps of $\Gamma_0(4m')$.) As all terms of (10) are non-negative, (i) follows by adding the four lower bounds. □

6.2. The sum $\sum r(n)^2/n$.

Lemma 6.3. *For $Y \ge 1$,*

$$S(Y) := \sum_{n\ge1} \frac{r(n)^2}{n}\, e^{-n/Y} = 2(\log Y)^2 + c_1 \log Y + c_2 + O(Y^{-1/2}),$$

$$c_1 = 4\gamma + 8\frac{L'}{L}(1,\chi) - 8\frac{\zeta'}{\zeta}(2) + \frac43 \log 2 = 9.758\ldots,$$

with an absolute constant c_2 ($c_2 = 6.12\ldots$ numerically); in particular $S(Y) \ge 2(\log Y)^2 - C$ for $Y \ge 1$. Moreover $\sum_{n\ge1} r(n)^2 e^{-n/X} = 4X\log X + O(X)$, so that $\sum_{n\le X} r(n)^2 \ll X \log 2X$, for $X \ge 1$.

Proof. Since $r(n)/4 = \sum_{d|n}\chi(d)$ is multiplicative with $r(p^k)/4 = k+1$, $\frac12(1+(-1)^k)$, 1 for $p \equiv 1,3 \pmod 4$ and $p=2$, the series $\sum_k (r(p^k)/4)^2 x^k$ equals $(1+x)(1-x)^{-3}$, $(1-x^2)^{-1}$, $(1-x)^{-1}$ respectively, and

$$D(s) = \sum_{n\ge1} \frac{r(n)^2}{n^s} = \frac{16\,\zeta(s)^2 L(s,\chi)^2}{(1+2^{-s})\,\zeta(2s)} \qquad (\mathrm{Re}\, s > 1).$$

By Mellin inversion, $S(Y) = \frac{1}{2\pi i}\int_{(1)} \Gamma(w) Y^w D(1+w)\,\mathrm{d}w$. We move the contour to $\mathrm{Re}\, w = -\frac12$: there $\zeta(2+2w) \neq 0$ and $1/\zeta(2+2w) \ll \log(|w|+2)$, $\zeta(1+w)$ and $L(1+w,\chi)$ are $\ll (|w|+2)^{1/4}$, and $\Gamma(w)$ decays exponentially, so the integral on the new line is $O(Y^{-1/2})$. The only pole crossed is $w=0$, of order three. With $\Gamma(w) = w^{-1}(1-\gamma w + O(w^2))$, $Y^w = 1 + w\log Y + \frac12 w^2(\log Y)^2 + O(w^3)$ and $D(1+w) = 4w^{-2}(1 + a_1 w + O(w^2))$, where $16\cdot\frac23\cdot L(1,\chi)^2/\zeta(2) = 16\cdot\frac23\cdot\frac{\pi^2}{16}\cdot\frac{6}{\pi^2} = 4$ and $a_1 = 2\gamma + 2\frac{L'}{L}(1,\chi) - 2\frac{\zeta'}{\zeta}(2) + \frac13\log 2$, the residue is $2(\log Y)^2 + 4(a_1-\gamma)\log Y + c_2$. The numerical value uses $L'(1,\chi)/L(1,\chi) = \gamma + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac14) = 0.2456\ldots$ (the case $\mathbb{Q}(i)$ of the Chowla–Selberg formula; it also follows from the functional equation of $L(s,\chi)$, $\zeta'(0,a) = \log\Gamma(a) - \frac12\log 2\pi$ for the Hurwitz zeta function and $\Gamma(\frac14)\Gamma(\frac34) = \pi\sqrt{2}$) and $\zeta'(2)/\zeta(2) = -0.5700\ldots$. Since $c_1 > 0$, $S(Y) \ge 2(\log Y)^2 + c_2 - O(1)$ for $Y \ge 1$. The last assertion is the same argument for $\frac{1}{2\pi i}\int \Gamma(w) X^w D(w)\mathrm{d}w$ with the contour moved to $\mathrm{Re}\, w = \frac34$, where only the double pole at $w=1$ is crossed, and $\sum_{n\le X} r(n)^2 \le e\sum_n r(n)^2 e^{-n/X}$. □

6.3. **The main term for small n.**

Lemma 6.4. *Let $Q = Q_{m,m'}$ and $n \ge 1$. Then $\beta_2(n) \le 2$; $\beta_p(n) \le \sigma_{-1}(p^{v_p(n)})$ for $p \nmid 2mm'$; and $\beta_p(n) \le 7\,(\min(v_p(n), v_p(mm')) + 1)$ for $p \mid mm'$, while $\beta_p(n) \le 1 + 1/p$ for $p \mid mm'$, $p \nmid n$. Consequently, for $Y \ge 1$,*

$$\sum_{n\ge1} \frac{r(n)\,M(m,m';n)}{n}\, e^{-n/Y} \;\ll\; \frac{Y}{mm'}\,\log(2Y)\,(\log\log 3mm')^C$$

with an absolute constant C.

Proof. For a prime p, $a \ge 1$ and $t \in \mathbb{Z}$ let $N_2(t;p^a) = \#\{(x,y) \bmod p^a :\ x^2+y^2 \equiv t\}$. We claim

$$N_2(t;p^a) \le 2\,(\min(v_p(t),a)+1)\,p^a \quad (p \text{ odd}), \qquad N_2(t;2^a) \le 2^{a+1}. \tag{11}$$

For odd p, the solutions with $(x,y) \not\equiv (0,0) \pmod p$ are non-singular and lift from p to p^a with multiplicity p^{a-1}; modulo p there are $p - \chi(p) \le 2p$ of them if $p \nmid t$ and $2(p-1)$ or 0 if $p \mid t$. The solutions with $x \equiv y \equiv 0 \pmod p$ exist only if $p^2 \mid t$ and are $p^2 N_2(t/p^2; p^{a-2})$ in number for $a \ge 2$. Hence $N_2(t;p^a) \le 2p^a + p^2 N_2(t/p^2;p^{a-2})$, and (11) follows by induction on a ($N_2(t;p) \le 2p$, $N_2(t;1) = 1$). For $p=2$ and $a \ge 3$: if $t \equiv 1 \pmod 4$, exactly one of x,y is odd, and since the odd squares modulo 2^a are the residues $\equiv 1 \pmod 8$, each taken four times, while $y^2 \equiv 0$ or $4 \pmod 8$ according as $4 \mid y$ or not, there are $2\cdot 4\cdot 2^{a-2} = 2^{a+1}$ solutions; if $t \equiv 2 \pmod 8$ both are odd and there are $4\cdot 4\cdot 2^{a-3} = 2^{a+1}$ solutions; if $t \equiv 3 \pmod 4$ or $t \equiv 6 \pmod 8$ there are none; and if $4 \mid t$ both are even and $N_2(t;2^a) = 4N_2(t/4;2^{a-2})$, so (11) follows by induction from $N_2(t;2^a) \le 2^{a+1}$ for $a \le 2$, which one checks by hand.

Now $\beta_2(n) = \lim_a 2^{-3a}\sum_{(z,w) \bmod 2^a} N_2\big((n - m'(z^2+w^2))\bar m; 2^a\big) \le \lim_a 2^{-3a}\cdot 2^{2a}\cdot 2^{a+1} = 2$, where $\bar m$ is an inverse of m modulo 2^a. For $p \nmid 2mm'$ the formula $\beta_p(n) = (1-p^{-2})\sigma_{-1}(p^{v_p(n)})$ is (6) of Part I. Let $p \mid mm'$, say $p^e \,\|\, m$ with $e \ge 1$, $p \nmid m'$, and $p^k \,\|\, n$. By (11), with $X = x^2+y^2$,

$$\beta_p(n) = \lim_a p^{-3a} \sum_{(x,y) \bmod p^a} N_2\big((n - mX)\bar m'; p^a\big) \le 2\Big(1 + \sum_{i\ge1} \mathrm{Prob}_{(x,y)}\big(p^i \mid n - mX\big)\Big),$$

the probability referring to (x,y) uniform modulo p^a with $a \ge i$. For $i \le \min(k,e)$ the probability is 1. If $k < e$ it is 0 for $i > k$. If $k \ge e$ and $i > e$, then $p^i \mid n - mX$ if and only if $X \equiv n_1 \pmod{p^{i-e}}$ with $n_1 = (n/p^e)(m/p^e)^{-1}$, which has probability $N_2(n_1; p^{i-e})p^{-2(i-e)} \le 2(i-e+1)p^{-(i-e)}$ by (11), and $\sum_{j\ge1} 2(j+1)p^{-j} \le \frac{5}{2}$ for $p \ge 3$. Hence $\beta_p(n) \le 2(\min(k,e) + \frac{7}{2}) \le 7(\min(k,e)+1)$; for $k = 0$ the exact value $1 - \chi(p)/p$ is (7) of Part I.

For the sum, write $M(m,m';n) = \pi^2 n\,\beta(n)/(mm')$ with $\beta(n) = \prod_p \beta_p(n)$, so that the sum is $\frac{\pi^2}{mm'}\sum_n r(n)\beta(n)e^{-n/Y}$, and $\beta(n) \le 2\,\sigma_{-1}(n)\prod_{p\mid mm'}(1+\frac{1}{p})\prod_{p\mid(n,mm')} 7(v_p(n)+1)$. Write $n = gk$ with g composed of the primes of mm' and $(k, mm') = 1$; then $r(n) \le 4d(g)d(k)$, $\sigma_{-1}(n) \le 2^{\omega(g)}\sigma_{-1}(k)$, and $\sum_{k\le x} d(k)\sigma_{-1}(k) \ll x\log 2x$ (write $\sigma_{-1}(k) = \sum_{u\mid k} 1/u$ and sum over u first). Therefore

$$\sum_n r(n)\beta(n)e^{-n/Y} \;\ll\; \prod_{p\mid mm'}\Big(1+\frac{1}{p}\Big)\sum_g \frac{7^{\omega(g)}\,2^{\omega(g)}\,d(g)^2}{g}\,Y\log(2Y) \;\ll\; Y\log(2Y)\prod_{p\mid mm'}\Big(1+\frac{C}{p}\Big),$$

since $\sum_{j\ge1} 14(j+1)^2p^{-j} \le C/p$ for $p \ge 3$, and $\prod_{p\mid mm'}(1+C/p) \le \exp(C\sum_{p\mid mm'} 1/p) \ll (\log\log 3mm')^C$. □

Proof of Theorem 1.5(iii). By (ii) we may replace (m,m') by $(1,mm')$, and we write $n = mm'$. Apply (i) to the pair $(1,n)$ and keep only the two cusps $Q \in \{n, 4n\}$ (of widths n and $4n$), with $Y = Q/(4\pi)$, so that $n/(4\pi) \le Y \le n/\pi$; we may assume $n \ge 4\pi$. Keep only $n' < n$ in the inner sum of (i) (we write n' for the summation variable there), where $a(n) = r(n) - M(1,mm';n)$ and $M \ge 0$, so that $|a(n)|^2 \ge r(n)^2 - 2r(n)M(1,mm';n)$:

$$\sum_{n\ge1}\frac{|a(n)|^2}{n}e^{-n/Y} \;\ge\; S(Y) - \sum_{n\ge mm'}\frac{r(n)^2}{n}e^{-n/Y} - 2\sum_{n\ge1}\frac{r(n)M(1,mm';n)}{n}e^{-n/Y}.$$

By Lemma 6.3, $S(Y) \ge 2(\log Y)^2 - C$, and the tail is at most

$$\sum_{j\ge0} e^{-\pi 2^j}\,\frac{1}{2^j mm'}\sum_{n<2^{j+1}mm'} r(n)^2 \;\ll\; \sum_{j\ge0} e^{-\pi 2^j}\log(2^{j+2}mm') \;\ll\; \log(2mm').$$

By Lemma 6.4 with (m,m') replaced by $(1,n)$, the last sum is at most a constant times $(Y/n)\log(2Y)(\log\log 3n)^C \le \log(2n)(\log\log 3n)^C$. Summing over the two Q and using

$$2\big(\log(n/4\pi)\big)^2 + 2\big(\log(n/\pi)\big)^2 \;\ge\; 4(\log n)^2 - 4\log(4\pi^2)\log n,$$

(i) gives

$$\|G_{m,m'}\|^2 = \|G_{1,n}\|^2 \;\ge\; \frac{1}{\pi}(\log n)^2 - \frac{\log(4\pi^2)}{\pi}\log n - C\log(2n)(\log\log 3n)^C,$$

which is (iii). □

The same computation at the four cusps of the pair (m,m') itself gives $\frac{1}{\pi}((\log m)^2 + (\log m')^2)$ in place of $\frac{1}{\pi}(\log mm')^2$: the cusp sums of the fundamental domain of (m,m') depend on the factorisation, their total does not, and the pair $(1,mm')$ gives the best lower bound.

6.4. **The matching upper bound.** The proof of Theorem 1.2 of Part I loses the factor $(mm')^\varepsilon$ at two places only: the representation numbers $r(S,n)$ of the forms $S = A(x_1^2+x_2^2) + B(x_3^2+x_4^2)$ at the cusps are bounded by $n^\varepsilon(1 + g_0 n/mm')$ through $r(\alpha)r(\beta) \le 16\,d(\alpha)d(\beta)$, and the genus terms through Waibel's general bound $r(\mathrm{gen}S, n) \ll n(n,L)^{1/2}(nL)^\varepsilon/\sqrt{\det S}$. Replacing the first by the mean value of $r(\alpha)^2r(\beta)^2$ (Lemma 6.3) and the second by the local densities themselves gives the true order.

Theorem 6.5 (The exact order of the norm)**.** *There is an absolute constant C such that for all coprime odd positive integers m, m'*

$$\|G_{m,m'}\|^2 \;\le\; C\,(\log 3mm')^2\,(\log\log 3mm')^C,$$

and for every k a constant C_k such that $\|G_{m,m'}\|^2 \le C_k(\log 3mm')^2$ whenever $\omega(mm') \le k$. Consequently $\|G_{m,m'}\|^2 \asymp_k (\log mm')^2$ for $mm' \to \infty$ with $\omega(mm') \le k$, and Theorem 1.1(ii) holds with $(mm')^\varepsilon$ replaced by $(\log 3mm')(\log\log 3mm')^C$.

Proof. We follow the proof of Theorem 1.2 of Part I (Section 2.5 there) and keep its notation: $L = 4mm'$; for a class of pairs (a,c) with $(c,L) = d < L$ we have $\hat d \in \{d, d/2\}$, the form S with $A = \kappa(m'/d_{m'})d_m$, $B = \kappa(m/d_m)d_{m'}$, $\kappa \in \{\frac14, \frac12, 1, 2, 4, 8\}$, $AB = \kappa^2 mm'$, $g := (A,B) = \kappa g_0 \le 8d$ (Lemma 2.11 there), and the class contributes at most

$$\frac{256}{\pi}\sum_{n\ge1}\frac{r(S,n)^2 + r(\mathrm{gen}S,n)^2}{n}\,e^{-\lambda n}, \qquad \lambda = \lambda_d = \frac{2\sqrt3\pi\hat d}{L}, \tag{12}$$

to the sum of Lemma 2.8 there, while the class $d = L$ (the cusp ∞) contributes $O(1)$, because $|a_G(n)| \le T(m,m';n) + M(m,m';n) \ll n^3$ trivially and the exponential is $e^{-2\sqrt3\pi n}$. We use throughout

$$\begin{gathered}\lambda AB = \frac{\sqrt3\pi}{2}\,\kappa^2\hat d \;\ge\; c_0\hat d, \qquad c_0 = \frac{\sqrt3\pi}{32}, \qquad \frac{g}{\lambda AB} \le \frac{16}{c_0},\\ Y_A := \frac{1}{\lambda A} \le \frac{3m}{d_m^2}, \qquad Y_B := \frac{1}{\lambda B} \le \frac{3m'}{d_{m'}^2},\end{gathered} \tag{13}$$

the last because $L/(\hat dA) \le 2L/(dA) = 8m/(d_2\kappa d_m^2) \le 32m/d_m^2$ with $d = d_2d_md_{m'}$, $d_2 \mid 4$. From Lemma 6.3 and partial summation, for $Y > 0$,

$$\begin{gathered}\sum_{\alpha\ge1}\frac{r(\alpha)^2}{\alpha}e^{-\alpha/Y} \ll (1+\log^+ Y)^2, \qquad \sum_{\alpha\ge1} r(\alpha)^2 e^{-\alpha/Y} \ll Y(1+\log^+ Y),\\ \sum_{\alpha\ge1}\frac{r(\alpha)^2}{\alpha^{1/2}}e^{-\alpha/Y} \ll Y^{1/2}(1+\log^+ Y)\end{gathered} \tag{14}$$

(for $Y < 1$ the sums are $\ll e^{-1/(2Y)}$).

The representation numbers. $r(S,n) = \sum_{A\alpha+B\beta=n} r(\alpha)r(\beta)$ has at most $1 + gn/(AB)$ terms, so by Cauchy–Schwarz $r(S,n)^2 \le (1+gn/(AB))\sum_{A\alpha+B\beta=n} r(\alpha)^2r(\beta)^2$, and

$$\sum_n \frac{r(S,n)^2}{n}e^{-\lambda n} \le \sum_{(\alpha,\beta)\ne(0,0)} r(\alpha)^2r(\beta)^2e^{-\lambda(A\alpha+B\beta)}\Big(\frac{1}{A\alpha+B\beta} + \frac{g}{AB}\Big).$$

The terms with $\beta = 0$ give, by (14), $\frac1A\sum_\alpha r(\alpha)^2\alpha^{-1}e^{-\lambda A\alpha} + \frac{g}{AB}\sum_\alpha r(\alpha)^2e^{-\lambda A\alpha} \ll \frac1A(1+\log^+ Y_A)^2 + \frac{g}{\lambda A^2B}(1+\log^+ Y_A)$, and likewise for $\alpha = 0$. For $\alpha,\beta \ge 1$ we use $A\alpha + B\beta \ge 2\sqrt{AB\alpha\beta}$ and (14):

$$\begin{aligned}\sum_{\alpha,\beta\ge1}\frac{r(\alpha)^2r(\beta)^2e^{-\lambda(A\alpha+B\beta)}}{A\alpha+B\beta} &\le \frac{1}{2\sqrt{AB}}\sum_\alpha\frac{r(\alpha)^2e^{-\lambda A\alpha}}{\alpha^{1/2}}\sum_\beta\frac{r(\beta)^2e^{-\lambda B\beta}}{\beta^{1/2}}\\ &\ll \frac{(1+\log^+ Y_A)(1+\log^+ Y_B)}{\lambda AB},\end{aligned}$$

and $\frac{g}{AB}\sum_{\alpha,\beta\ge1} r(\alpha)^2 r(\beta)^2 e^{-\lambda(A\alpha+B\beta)} \ll \frac{g}{\lambda^2A^2B^2}(1+\log^+ Y_A)(1+\log^+ Y_B)$. With (13), $\log^+ Y_A \le \log(3m)$, $\log^+ Y_B \le \log(3m')$, the whole sum is

$$\sum_n \frac{r(S,n)^2}{n} e^{-\lambda n} \ll \frac{(1+\log^+(3m/d_m^2))^2}{A} + \frac{(1+\log^+(3m'/d_{m'}^2))^2}{B} + \frac{(\log 3mm')^2}{\hat d}. \tag{15}$$

The genus terms. By Siegel's theorem, $r(\mathrm{gen}S,n) = \pi^2 n\,\beta_S(n)/(AB)$ with $\beta_S(n) = \prod_p \beta_p(S;n)$, since $\sqrt{\det S} = 4AB$. If $p \mid g$ then $S = pS_1$ with S_1 integral, and counting solutions modulo p^a gives $\beta_p(S;n) = p\,\beta_p(S_1;n/p)$ if $p \mid n$ and 0 otherwise; iterating over the prime factors of g (including $p=2$ when κ is even) gives $\beta_S(n) = g\,\beta_{S'}(n/g)$ if $g \mid n$ and 0 otherwise, where $S' = S/g = Q_{A',B'}$ with $A' = A/g$, $B' = B/g$ coprime and odd. For S', Lemma 6.4 gives $\beta_2(S';n) \le 2$, $\beta_p(S';n) \le \sigma_{-1}(p^{v_p(n)})$ for $p \nmid 2A'B'$, $\beta_p(S';n) \le 1+1/p$ for $p \mid A'B'$, $p \nmid n$, and $\beta_p(S';n) \le 7(v_p(n)+1)$ for $p \mid (n, A'B')$. Hence $\beta_{S'}(n)^2 \le 4\,(\sigma(A'B')/A'B')^2 f(n)$ with the multiplicative function f defined by $f(p^k) = \sigma_{-1}(p^k)^2$ for $p \nmid A'B'$ and $f(p^k) = 49(k+1)^2$ for $p \mid A'B'$, $k \ge 1$. Since $f(p^k) \le 49(k+1)^2$ for all p and $f(p) = 1 + O(1/p)$ for $p \nmid A'B'$, the mean-value inequality for non-negative multiplicative functions [12, Theorem 01] gives

$$\sum_{n\le X} f(n) \ll \frac{X}{\log X}\prod_{p\le X}\Big(\sum_{k\ge0}\frac{f(p^k)}{p^k}\Big) \ll X\prod_{p\mid A'B'}\Big(1+\frac{200}{p}\Big) \ll X(\log\log 3A'B')^{200} \qquad (X\ge2),$$

so that $\sum_{n\le X}\beta_S(n)^2 = g^2\sum_{n'\le X/g}\beta_{S'}(n')^2 \ll gX(\log\log 3mm')^C$, and by partial summation $\sum_n n\beta_S(n)^2 e^{-\lambda n} \ll g\lambda^{-2}(\log\log 3mm')^C$. Therefore, by (13),

$$\sum_n \frac{r(\mathrm{gen}S,n)^2}{n} e^{-\lambda n} = \frac{\pi^4}{(AB)^2}\sum_n n\beta_S(n)^2 e^{-\lambda n} \ll \frac{g\,(\log\log 3mm')^C}{(\lambda AB)^2} \ll \frac{(\log\log 3mm')^C}{\hat d}. \tag{16}$$

The sum over the classes. The classes are indexed by $d_2 \mid 4$, $d_m \mid m$, $d_{m'} \mid m'$ (with κ and $\hat d$ determined by the class), and $\hat d \ge d/2$. Inserting (15) and (16) into (12) and summing,

$$\|G_{m,m'}\|^2 \ll 1 + \Big((\log 3mm')^2 + (\log\log 3mm')^C\Big)\sum_{d\mid L}\frac{2}{d}$$
$$+(\log 3m)^2\sum_{\text{classes}}\frac{1}{A} + (\log 3m')^2\sum_{\text{classes}}\frac{1}{B},$$
$$\text{where}\quad \sum_{d\mid L}\frac{1}{d} = \frac{\sigma(L)}{L} \ll \log\log 3L,$$
$$\sum_{\text{classes}}\frac{1}{A} \le \sum_{d_2\mid4}\sum_{d_m\mid m}\sum_{d_{m'}\mid m'}\frac{4d_{m'}}{m'd_m} = 12\,\frac{\sigma(m)}{m}\,\frac{\sigma(m')}{m'},$$

the last being $\ll (\log\log 3mm')^2$, and likewise for B. This proves the first bound. If $\omega(mm') \le k$, then $\sigma(mm')/mm' \le 2^k$, $\sigma(L)/L \le 2^{k+2}$ and $\prod_{p\mid A'B'}(1+200/p) \le 201^k$, so all the log log factors are $\le C_k$, which gives the second bound; the lower bound is Theorem 1.5, and the last assertion is Part I, Lemma 2.2 and Theorem 2.3, where the norm enters only through the Cauchy–Schwarz step, which costs exactly the factor $\|G_{m,m'}\|$. □

Corollary 6.6 (The spectral second moment)**.** *Let $\mathcal{B}^*$ be an orthogonal basis of $S_2(\Gamma_0(4mm'))$ of eigenforms of the Hecke operators T_n, $(n, 2mm') = 1$, as in Part I, Section 5.5. Then*

$$\sum_{\varphi\in\mathcal{B}^*}\frac{|\langle\Theta_{m,m'},\varphi\rangle|^2}{\|\varphi\|^2} = \|G_{m,m'}\|^2 \asymp_k (\log mm')^2 \qquad (\omega(mm')\le k),$$

and, for every N with $(N,2mm')=1$, Theorem 1.1(ii) is the bound $|\sum_\varphi \langle\Theta,\varphi\rangle a_\varphi(N)/\|\varphi\|^2| \ll d(N)N^{1/2}(\log 3mm')(\log\log 3mm')^C$ for the spectral sum (29) of Part I.

The first identity is Parseval's, since $E \perp \varphi$. Part I, Section 5.5, described the Rankin–Selberg structure of the coefficients, $\langle\Theta_{m,m'},\varphi\rangle = \mathcal{E}_\varphi(m,m')L(\frac12,f)L(\frac12,f\otimes\chi_{-4})$ with a factor $\mathcal{E}_\varphi$ built from local data, without computing it. We now compute it: for newforms the local data cancel completely.

6.5. **The Rankin–Selberg formula.** We write $L(s,\varphi)=\sum_k a_\varphi(k)k^{-s}$ and $L(s,\varphi\otimes\chi)=\sum_k \chi(k)a_\varphi(k)k^{-s}$ for a newform $\varphi=\sum a_\varphi(k)q^k$ of weight 2 with $a_\varphi(1)=1$, so that $s=1$ is the centre of the critical strip, and $\eta_Q(\varphi)=\pm1$ for its Atkin–Lehner eigenvalues, $\varphi|_2W_Q=\eta_Q(\varphi)\varphi$ for $Q\,\|\,L$; recall $a_\varphi(p)=-\eta_p(\varphi)$ for $p\,\|\,L$.

Theorem 6.7 (The Rankin–Selberg formula)**.** *Let n be an odd square-free positive integer and $L=4n$.*

(i) *For a newform φ of weight 2 and level L,*

$$\langle\Theta_{1,n},\varphi\rangle = \frac{4}{\pi^2}\,\chi(n)\,\eta_n(\varphi)\,L(1,\varphi)\,L(1,\varphi\otimes\chi).$$

(ii) *For a newform f of weight 2 and level $M \mid L$, $M=2^eM_1$ with M_1 odd, and $d \mid L/M$,*

$$\langle\Theta_{1,n},f(d\,\cdot)\rangle = \frac{4}{\pi^2}\,\chi(M_1)\,\eta_{M_1}(f)\,\frac{\prod_{p\mid n/M_1}(1+\chi(p))}{d}\,L(1,f)\,L(1,f\otimes\chi),$$

where $\eta_{M_1}(f)=\prod_{p\mid M_1}\eta_p(f)$; in particular $\langle\Theta_{1,n},f(d\,\cdot)\rangle=0$ unless every prime of n/M_1 is $\equiv 1 \pmod 4$, and then the factor is $2^{\omega(n/M_1)}/d$.

(iii) *For every factorisation $n=mm'$ into coprime factors,*

$$\langle\Theta_{m,m'},f(d\,\cdot)\rangle = \chi(m)\,\eta_{(m,M_1)}(f)\,\langle\Theta_{1,n},f(d\,\cdot)\rangle;$$

for newforms of level L this reads $\langle\Theta_{m,m'},\varphi\rangle = \frac{4}{\pi^2}\chi(m')\eta_{m'}(\varphi)L(1,\varphi)L(1,\varphi\otimes\chi)$. In particular $|\langle\Theta_{m,m'},f(d\,\cdot)\rangle|$ does not depend on the factorisation.

Here (i) is the case $M=L$, $d=1$ of (ii), and it is proved first; (ii) needs in addition the action of the Atkin–Lehner matrices of level L on the oldforms $f(dz)$.

Lemma 6.8. *Let f be a newform of level $M \mid L$, $d \mid L/M$, $Q \mid n$, and put $Q_1=(Q,M)$, $Q_2=(Q,d)$, $Q_3=Q/(Q_1Q_2)$, $d'=dQ_3/Q_2$. Then $f(d\,\cdot)|_2W_Q^{(L)} = (d'/d)\,\eta_{Q_1}(f)\,f(d'\cdot)$.*

Proof. Since the odd part of L is square-free, Q_1,Q_2,Q_3 are pairwise coprime and $d' \mid L/M$. With $W_Q=\left(\begin{smallmatrix} Qa & b \\ Lc & Q\delta\end{smallmatrix}\right)$, $Q^2a\delta - Lbc = Q$, one checks

$$B_dW_Q = Q_2\,Y\,B_{d'}, \qquad Y=\begin{pmatrix} Q_1Q_2a & db/Q_2 \\ (L/(dQ_3))\,c & Q_1Q_3\delta\end{pmatrix},$$

where Y is integral ($Q_2 \mid d$, and $dQ_3 \mid L$) with $\det Y = Q_1$, Q_1 divides its diagonal entries and M divides $L/(dQ_3)$ because $dQ_3 \mid L/M$ (including the 2-part, as $d \mid L/M$); so Y is an Atkin–Lehner matrix of level M and index Q_1, and $f|_2Y=\eta_{Q_1}(f)f$. As $f(d\,\cdot)=d^{-1}f|_2B_d$ and scalars act trivially, $f(d\,\cdot)|_2W_Q = d^{-1}f|_2(YB_{d'}) = d^{-1}\eta_{Q_1}(f)f|_2B_{d'} = (d'/d)\eta_{Q_1}(f)f(d'\cdot)$. □

Proof of Theorem 6.7. Reduction to $(1,n)$. By Lemma 6.1, applied to each prime of m in turn, $\Theta_{m,m'}|_2W_m=\chi(m)\Theta_{1,n}$, and $|_2W_m$ is an isometry, so $\langle\Theta_{m,m'},f(d\,\cdot)\rangle = \chi(m)\langle\Theta_{1,n},f(d\,\cdot)|_2W_m\rangle = \chi(m)(d'/d)\eta_{(m,M_1)}(f)\langle\Theta_{1,n},f(d'\cdot)\rangle$ by Lemma 6.8 (with $Q=m$), and (ii) for d and d' gives (iii); for $M=L$, $\chi(m)\chi(n)=\chi(m')$ and $\eta_m\eta_n=\eta_{m'}$.

The Eisenstein series. For $\operatorname{Re} s > \frac12$ let

$$E(z,s) = \sum_{\gamma\in\Gamma_\infty\backslash\Gamma_0(4)} \chi(d_\gamma)\, j(\gamma,z)^{-1}\, \operatorname{Im}(\gamma z)^s, \qquad j(\gamma,z)=cz+d, \quad \gamma=\left(\begin{smallmatrix} a & b\\ c & d\end{smallmatrix}\right),$$

where $\Gamma_\infty = \{\pm\left(\begin{smallmatrix}1 & t\\ 0 & 1\end{smallmatrix}\right)\}$; the terms for γ and $-\gamma$ agree because $\chi(-1)=-1$. The series converges absolutely, $E(\gamma z,s)=\chi(d_\gamma)j(\gamma,z)E(z,s)$ for $\gamma\in\Gamma_0(4)$, and by Hecke's method ([6, Chapter 4]) $E(z,s)$ continues analytically to a neighbourhood of $s=0$, where $E(z,0)$ is a holomorphic modular form in $M_1(\Gamma_0(4),\chi)$; its constant term is 1, because the second term of the constant term carries the factor $\sum_{d \bmod c,\,(d,c)=1}\chi(d)=0$ for $4\mid c$. Since $\dim M_1(\Gamma_0(4),\chi)=1$ and θ^2 has constant term 1, $E(z,0)=\theta^2(z)$. Put $g(z)=\theta^2(nz)\in M_1(\Gamma_0(L),\chi)$ and

$$I(s) = \int_{\Gamma_0(L)\backslash\mathbb{H}} E(z,s)\, g(z)\overline{\varphi(z)}\, y^2\, \mathrm{d}\mu, \qquad \mathrm{d}\mu = \frac{\mathrm{d}x\,\mathrm{d}y}{y^2},$$

which is analytic near $s=0$ with $I(0)=\langle\Theta_{1,n},\varphi\rangle$ (the integrand is $\Gamma_0(L)$-invariant, $E(z,s)$ has polynomial growth at the cusps uniformly for s near 0, and φ decays exponentially).

Decomposition over the cusps. The map $\Gamma_\infty\gamma\mapsto\gamma^{-1}\infty$ identifies $\Gamma_\infty\backslash\Gamma_0(4)$ with the $\Gamma_0(4)$-orbit of ∞ in $\mathbb{P}^1(\mathbb{Q})$, and the right action of $\Gamma_0(L)$ decomposes this orbit into the $\Gamma_0(L)$-classes of cusps lying over ∞, which are the cusps a/c with $4\mid c$, i.e. the classes of denominators $4n/Q$, $Q\mid n$, represented by $\mathfrak{a}_Q = W_Q\infty$. For each Q let $\varrho_Q=\left(\begin{smallmatrix} a & b\\ (L/Q)c_1 & Qd\end{smallmatrix}\right)\in\mathrm{SL}_2(\mathbb{Z})$ with $W_Q=\varrho_Q B_Q$ as in the proof of Lemma 6.2, $\beta_Q=\varrho_Q^{-1}$, so that $\beta_Q^{-1}\infty=\mathfrak{a}_Q$ and $d_{\beta_Q}=a$. Then $\Gamma_\infty\backslash\Gamma_0(4)=\bigsqcup_{Q\mid n}\Gamma_\infty\backslash\Gamma_\infty\beta_Q\Gamma_0(L)$, and $\gamma'\mapsto\Gamma_\infty\beta_Q\gamma'$ identifies $\Gamma_{\mathfrak{a}_Q}\backslash\Gamma_0(L)$ with the Q-th piece, $\Gamma_{\mathfrak{a}_Q}=\beta_Q^{-1}\Gamma_\infty\beta_Q\cap\Gamma_0(L)$ being the stabiliser of the cusp, of width $w_Q=Q$. Writing $H(z)=g(z)\overline{\varphi(z)}y^2$, the automorphy of g and φ gives $H(z)=\chi(d_{\gamma'})^{-1}j(\gamma',z)H(\gamma' z)$ for $\gamma'\in\Gamma_0(L)$, hence

$$\chi(d_{\beta_Q\gamma'})\, j(\beta_Q\gamma',z)^{-1}\operatorname{Im}(\beta_Q\gamma' z)^s H(z) = \chi(a)\, K_Q(\gamma' z), \quad K_Q(w)=j(\beta_Q,w)^{-1}\operatorname{Im}(\beta_Q w)^s H(w),$$

and unfolding, $I(s)=\sum_{Q\mid n}\chi(a)\int_{\Gamma_{\mathfrak{a}_Q}\backslash\mathbb{H}} K_Q\,\mathrm{d}\mu$. With the scaling matrix $\sigma_Q=\beta_Q^{-1}\left(\begin{smallmatrix} Q^{1/2} & 0\\ 0 & Q^{-1/2}\end{smallmatrix}\right)=Q^{-1/2}W_Q$ we have $\beta_Q\sigma_Q u=Qu$, $j(\beta_Q,\sigma_Q u)^{-1}=Q^{1/2}j(\sigma_Q,u)$, $\operatorname{Im}(\beta_Q\sigma_Q u)=Q\operatorname{Im}u$, and

$$H(\sigma_Q u) = (g|_1\sigma_Q)(u)\overline{(\varphi|_2\sigma_Q)(u)}\, j(\sigma_Q,u)\overline{j(\sigma_Q,u)}^{\,2}|j(\sigma_Q,u)|^{-4}(\operatorname{Im}u)^2,$$

so that all the factors $j(\sigma_Q,u)$ cancel and

$$\int_{\Gamma_{\mathfrak{a}_Q}\backslash\mathbb{H}} K_Q\,\mathrm{d}\mu = Q^{1/2+s}\int_0^\infty\!\!\int_0^1 (g|_1\sigma_Q)(u)\,\overline{(\varphi|_2\sigma_Q)(u)}\, v^s\,\mathrm{d}u_1\,\mathrm{d}v \qquad (u=u_1+iv).$$

Here $g|_1\sigma_Q=g|_1W_Q$ and $\varphi|_2\sigma_Q=\varphi|_2W_Q$, because scalar matrices act trivially; and, exactly as in the proof of Lemma 6.1, $B_QW_Q=Q\left(\begin{smallmatrix} Qa & b\\ 4(n/Q)c_1 & d\end{smallmatrix}\right)$ with the second factor in $\Gamma_0(4n/Q)$, so that $g|_1W_Q=Q^{-1/2}\chi(d)\,\theta^2((n/Q)z)$. Since $Qad-4(n/Q)bc_1=1$ gives $ad\equiv Q \pmod 4$, $\chi(a)\chi(d)=\chi(Q)$, and by Parseval, writing $\varphi|_2W_Q=\sum_k c_Q(k)e(kz)$,

$$I(s) = \sum_{Q\mid n}\chi(Q)\,\Gamma(s+1)(4\pi)^{-s-1}\sum_{j\ge1}\frac{r(j)\,\overline{c_Q((n/Q)j)}}{((n/Q)j)^{s+1}} \qquad (\operatorname{Re}s>\tfrac12). \tag{17}$$

For a newform φ of level L, $c_Q(k)=\eta_Q(\varphi)a_\varphi(k)$, with real a_φ.

The Dirichlet series. For $p\mid n/Q$ we have $p\mid L$, so $a_\varphi((n/Q)j)=a_\varphi(n/Q)a_\varphi(j)$ for all j, and $r(j)=4\sum_{d\mid j}\chi(d)$ is 4 times the coefficient of the Eisenstein series with Satake parameters

$(1,\chi(p))$; the standard computation of the Rankin–Selberg Euler product gives, for $\operatorname{Re} w > \frac{3}{2}$,

$$\sum_{j\geq 1}\frac{r(j)a_\varphi(j)}{j^w} = 4\,L(w,\varphi)L(w,\varphi\otimes\chi)\prod_{p\nmid L}\big(1-\chi(p)p^{1-2w}\big) = \frac{4\,L(w,\varphi)L(w,\varphi\otimes\chi)}{L(2w-1,\chi)\prod_{p|n}(1-\chi(p)p^{1-2w})}$$

where, for $p\nmid 2L$ with $a_\varphi(p)=\alpha+\beta$, $\alpha\beta=p$ and $x=p^{-w}$, the local factor of the left-hand side is

$$\sum_{j\geq 0}\sigma_\chi(p^j)a_\varphi(p^j)x^j = \frac{1-\chi(p)p\,x^2}{(1-\alpha x)(1-\beta x)(1-\chi(p)\alpha x)(1-\chi(p)\beta x)},$$

while at $p\mid L$, $a_\varphi(p^j)=a_\varphi(p)^j$ gives $(1-a_\varphi(p)x)^{-1}(1-\chi(p)a_\varphi(p)x)^{-1}$, the product of the two degenerate Euler factors, with no numerator. Both sides continue to $w=1$. Hence, letting $s\to 0$,

$$\langle\Theta_{1,n},\varphi\rangle = \frac{1}{\pi}\cdot\frac{L(1,\varphi)L(1,\varphi\otimes\chi)}{L(1,\chi)\prod_{p|n}(1-\chi(p)/p)}\sum_{Q|n}\chi(Q)\eta_Q(\varphi)\frac{a_\varphi(n/Q)}{n/Q}.$$

Finally $a_\varphi(n/Q)/(n/Q)=\prod_{p|n/Q}(-\eta_p(\varphi)/p)$ and $\eta_Q=\prod_{p|Q}\eta_p$, so the last sum is

$$\eta_n(\varphi)\sum_{Q|n}\chi(Q)\prod_{p|n/Q}\Big(-\frac{1}{p}\Big) = \eta_n(\varphi)\prod_{p|n}\Big(\chi(p)-\frac{1}{p}\Big) = \chi(n)\eta_n(\varphi)\prod_{p|n}\Big(1-\frac{\chi(p)}{p}\Big),$$

and $L(1,\chi)=\pi/4$. This proves (i).

The oldforms. Let now $\varphi=f(d\cdot)$ with f a newform of level $M=2^eM_1$, $d=d_2d_{\mathrm{o}}$ with $d_2\mid 2^{2-e}$ and d_{o} odd, and write $n=M_1d_{\mathrm{o}}R$ (the three factors are pairwise coprime, since n is square-free and $d_{\mathrm{o}}\mid n/M_1$). For $Q\mid n$ write $Q=Q_1Q_2Q_3$ with $Q_1\mid M_1$, $Q_2\mid d_{\mathrm{o}}$, $Q_3\mid R$. By Lemma 6.8, $c_Q(k)=(d'/d)\eta_{Q_1}(f)a_f(k/d')$ for $d'\mid k$ and 0 otherwise, with $d'=dQ_3/Q_2$. Put $Q'=n/Q$ and $\delta=d_{\mathrm{o}}/Q_2$, which divides Q'; then $d'\mid Q'j$ if and only if $d_2Q_3\mid j$, and for $j=d_2Q_3j_1$ we have $Q'j/d'=Q''j_1$ with $Q''=Q'/\delta=(M_1/Q_1)(R/Q_3)$, while $Q'j=Q'd_2Q_3j_1$. Hence the Q-th inner sum in (17) at $s=0$ is

$$\frac{d'}{d}\,\eta_{Q_1}(f)\,\frac{4}{Q'd_2Q_3}\sum_{j_1\geq 1}\frac{\sigma_\chi(d_2Q_3j_1)\,a_f(Q''j_1)}{j_1} = \frac{4\,\eta_{Q_1}(f)}{Q'Q_2d_2}\,D_Q,$$

where

$$D_Q := \sum_{j_1\geq 1}\frac{\sigma_\chi(d_2Q_3j_1)\,a_f(Q''j_1)}{j_1},$$

since $d'/(dd_2Q_3)=1/(Q_2d_2)$. The series D_Q (continued from $\operatorname{Re} w>\frac{3}{2}$ as before) is an Euler product whose local factors differ from those of $\sum\sigma_\chi(j)a_f(j)j^{-w}$ only at the primes of d_2Q_3 (where the argument of σ_χ carries an extra factor p) and at the primes of Q'' (where the argument of a_f does). With $x=p^{-w}$ and $L_p=L_p(w,f)L_p(w,f\otimes\chi)$: at $p=2$, $\sigma_\chi(2^k)=1$ and the factor is $\sum_k a_f(2^k)x^k=L_2(w,f)=L_2$; at $p\mid M_1$, $a_f(p^{k+v})=a_f(p)^{k+v}$ gives $a_f(p)^vL_p$ ($v=1$ for $p\mid M_1/Q_1$, $v=0$ otherwise); and at $p\nmid 2M$, with $a_f(p)=\alpha+\beta$, $\alpha\beta=p$,

$$\begin{aligned}
\sum_{k\geq 0}\sigma_\chi(p^k)a_f(p^k)x^k &= (1-\chi(p)px^2)\,L_p,\\
\sum_{k\geq 0}\sigma_\chi(p^{k+1})a_f(p^k)x^k &= \big(1+\chi(p)-\chi(p)a_f(p)x\big)\,L_p,\\
\sum_{k\geq 0}\sigma_\chi(p^k)a_f(p^{k+1})x^k &= \big(a_f(p)-(1+\chi(p))px\big)\,L_p
\end{aligned}$$

(the second from $\sigma_\chi(p^{k+1}) = \sigma_\chi(p^k) + \chi(p)^{k+1}$, the third from $a_f(p^{k+1}) = (\alpha^{k+2} - \beta^{k+2})/(\alpha - \beta)$ and $\sum_k \sigma_\chi(p^k)(\alpha x)^k = ((1-\alpha x)(1-\chi(p)\alpha x))^{-1}$). Therefore, at $w = 1$,

$$D_Q = \frac{4\,L(1,f)L(1,f\otimes\chi)}{\pi \prod_{p\mid M_1}(1-\chi(p)/p)} \prod_{p\mid M_1/Q_1} a_f(p) \prod_{p\mid Q_3} \frac{1+\chi(p)-\chi(p)a_f(p)/p}{1-\chi(p)/p} \prod_{p\mid R/Q_3} \frac{a_f(p)-1-\chi(p)}{1-\chi(p)/p}.$$

Inserting this into (17), with $\Gamma(1)(4\pi)^{-1}$ and $1/(Q'Q_2) = 1/((M_1/Q_1)\,d_{\mathrm{o}}\,(R/Q_3))$, the sum over Q factors over the three parts of Q:

$$\begin{aligned}\langle \Theta_{1,n}, f(d\,\cdot)\rangle = \frac{4\,L(1,f)L(1,f\otimes\chi)}{\pi^2\, d_2 d_{\mathrm{o}} \prod_{p\mid M_1}(1-\chi(p)/p)} \Big[\sum_{Q_1\mid M_1} \chi(Q_1)\eta_{Q_1}(f) \prod_{p\mid M_1/Q_1} \frac{a_f(p)}{p}\Big]\Big[\sum_{Q_2\mid d_{\mathrm{o}}} \chi(Q_2)\Big] \\ \times \Big[\sum_{Q_3\mid R} \chi(Q_3) \prod_{p\mid Q_3} \frac{1+\chi(p)-\chi(p)a_f(p)/p}{1-\chi(p)/p} \prod_{p\mid R/Q_3} \frac{(a_f(p)-1-\chi(p))/p}{1-\chi(p)/p}\Big].\end{aligned}$$

The first bracket is $\chi(M_1)\eta_{M_1}(f)\prod_{p\mid M_1}(1-\chi(p)/p)$ as in (i); the second is $\prod_{p\mid d_{\mathrm{o}}}(1+\chi(p))$; and the third is a product over $p \mid R$ of

$$\frac{\chi(p)\big(1+\chi(p)-\chi(p)a_f(p)/p\big) + \big(a_f(p)-1-\chi(p)\big)/p}{1-\chi(p)/p} = \frac{(1+\chi(p))(1-1/p)}{1-\chi(p)/p} = 1+\chi(p),$$

the terms with $a_f(p)$ cancelling. Since $n/M_1 = d_{\mathrm{o}}R$ and $d = d_2 d_{\mathrm{o}}$, this is (ii). □

Remark 6.9. (i) The Atkin–Lehner symmetries alone show where the coefficients can be non-zero: $\langle \Theta_{1,n}, \varphi\rangle = \langle \Theta_{1,n}|W_n, \varphi|W_n\rangle = \chi(n)\eta_n(\varphi)\langle \Theta_{1,n},\varphi\rangle$ (Lemma 6.1 with $(m,m') = (n,1)$), and $\Theta|_2 W_L = -\Theta$ (Section 6.6, the cusp 0) gives $\langle \Theta,\varphi\rangle = -\eta_L(\varphi)\langle\Theta,\varphi\rangle$; so $\langle\Theta_{1,n},\varphi\rangle \neq 0$ forces $\eta_n(\varphi) = \chi(n)$ and $\eta_4(\varphi) = -\chi(n)$, i.e. the root number $-\eta_L(\varphi)$ of φ is $+1$. The formula is consistent with this: when $\eta_n \neq \chi(n)$ the right-hand side vanishes because one of the two central values does (root number -1). (ii) The theorem was checked numerically (`test_rs_formula.py`, `test_rs_old.py`, outputs in `results/`): the T_3-eigencomponents of $G_{1,p} = T - M$ are isolated, their norms computed by Rankin–Selberg and the central values from the functional equation. For newforms: at the level 20 (where $G_{1,5} = \frac{8}{3} f_{20}$) the two sides agree to six decimals, 0.21693; at the levels 52, 116 and 148 (one, three and two newforms with non-zero component, the last two with Hecke eigenvalues in $\mathbb{Q}(\sqrt{17})$; all with $\eta_p = +1$ and root number $+1$) the ratios are 0.99975; 0.99997, 0.99878, 0.99760; 1.00054, 0.99932. For oldforms, the component $v = c_1 f + c_2 f(2\,\cdot) + c_4 f(4\,\cdot)$ of $G_{1,p}$ in the space of f is its orthogonal projection, so $\|v\|^2 = \langle\Theta, v\rangle = \frac{4}{\pi^2}\chi(p)\eta_p(f)L(1,f)L(1,f\otimes\chi)(c_1 + c_2/2 + c_4/4)$ by (ii); the ratios of the two sides are 1.00071 for f of level 11 in $G_{1,11}$ ($c_1 : c_2 : c_4 = 1:3:4$), 1.00184 for f of level 26 in $G_{1,13}$, 0.99873 for f of level 58 in $G_{1,29}$, and 1.00068, 1.00035 for the two forms of level 74 in $G_{1,37}$ ($c_1 : c_2 = 1:2$ in the level-$2p$ cases), within the accuracy of the Rankin–Selberg norms. At the levels 44 and 76 ($p \equiv 3 \pmod 4$) $G_{1,p}$ has no newform component, in accordance with (i) and the vanishing of $L(1,\varphi\otimes\chi)$ there.

Corollary 6.10 (The second moment over the newforms of all levels)**.** *Let n be odd and square-free. For a newform f of weight 2 and level $M = 2^e M_1 \mid 4n$ let $S(f)$ be the span of the $f(d\,\cdot)$, $d \mid 4n/M$, Γ_f the Gram matrix $(\langle f(d\,\cdot), f(d'\cdot)\rangle)_{d,d'}$ and $w = (1/d)_d$. Then*

$$\|G_{1,n}\|^2 = \sum_f \big\|\mathrm{pr}_{S(f)}\Theta_{1,n}\big\|^2 = \frac{16}{\pi^4}\sum_f \gamma_f\, 4^{\omega(n/M_1)}\, L(1,f)^2 L(1,f\otimes\chi)^2, \qquad \gamma_f = w^{\mathrm{T}}\Gamma_f^{-1} w,$$

the sum over the newforms f of weight 2 and level dividing $4n$ for which every prime of n/M_1 is $\equiv 1 \pmod 4$. In particular

$$\sum_{\varphi\in H_2^{\mathrm{new}}(4n)} \frac{L(1,\varphi)^2 L(1,\varphi\otimes\chi)^2}{\langle\varphi,\varphi\rangle} \;\le\; \frac{\pi^4}{16}\,\|G_{1,n}\|^2 \;\ll\; (\log 3n)^2(\log\log 3n)^C,$$

and, since $\langle\varphi,\varphi\rangle \asymp n^{1+o(1)}$ for newforms of level $4n$, the harmonic second moment of the product $L(\frac12,\varphi)L(\frac12,\varphi\otimes\chi_{-4})$ of central values over the newforms of level $4n$ is $\ll (\log n)^{2+o(1)}$.

Proof. The spaces $S(f)$ are mutually orthogonal and span $S_2(\Gamma_0(4n))$; $\mathrm{pr}_{S(f)}\Theta = \sum_d c_d f(d\cdot)$ with $\Gamma_f c = v$, $v_d = \langle\Theta, f(d\cdot)\rangle = \kappa_f/d$ by Theorem 6.7(ii), where

$$\kappa_f = \tfrac{4}{\pi^2}\chi(M_1)\eta_{M_1}(f)2^{\omega(n/M_1)}L(1,f)L(1,f\otimes\chi);$$

hence $\|\mathrm{pr}_{S(f)}\Theta\|^2 = v^{\mathrm T}\Gamma_f^{-1}v = \kappa_f^2\gamma_f$. The Gram matrices are explicit (see [13, Section 2]); for a newform of level $4n$, $\gamma_f = 1/\langle f,f\rangle$. □

The two-sided bound of Corollary 6.6 is therefore a statement about the second moment of $L(\frac12,f)L(\frac12,f\otimes\chi_{-4})$ over the newforms f of all levels 2^eM_1 with $M_1 \mid n$, each weighted by the explicit $\gamma_f 4^{\omega(n/M_1)}$, with both the upper and the lower bound obtained from the geometry of the cusps rather than from the Petersson formula.

6.6. **All cusps, numerically.** For square-free m, m' the computation of Lemma 6.2 can be carried out at every cusp. The cusps of $\Gamma_0(L)$ are then in bijection with the divisors d of L (one cusp of denominator d for each d, since $(d, L/d) \le 2$), and we write $d = d_2\delta$ with $d_2 \mid 4$, $\delta \mid mm'$, $\delta_m = (\delta, m)$, $\delta_{m'} = (\delta, m')$, and

$$A = \delta_m\,\frac{m'}{\delta_{m'}}, \qquad B = \delta_{m'}\,\frac{m}{\delta_m}, \qquad AB = mm', \quad (A,B) = 1.$$

The cusps with $d_2 \in \{1,4\}$ are the images of ∞ under the Atkin–Lehner involutions $W_{L/d}$, their widths are $L/d = 4mm'/\delta$ and mm'/δ, and the proof of Lemma 6.1, applied prime by prime, gives $G_{m,m'}|_2 W_{L/d} = \varepsilon\, G_{A,B}$ with $|\varepsilon| = 1$ (the odd part of L/d moves the primes of δ_m from the first block to the second and those of $\delta_{m'}$ from the second to the first; the 2-part contributes a sign). The two cusps of Theorem 1.5 with the pair $(1, mm')$ are $\delta \in \{m, m'\}$. At the cusp with $d_2 = 2$, the function θ^2 is transformed into θ_2^2, $\theta_2(z) = \sum_{x\ \mathrm{odd}} e(x^2z/4)$ (from $\theta^2|_1 \left(\begin{smallmatrix}1&0\\2&1\end{smallmatrix}\right) = \theta_2^2$, which follows from the transformation formulae of $\theta_3, \theta_4, \theta_2$ under $z \mapsto -1/(4z)$), and $\Theta_{m,m'}$ into $(mm'/\delta)^{-1}\sum_n T^{\mathrm{odd}}(A,B;n)e(\delta nz/(4mm'))$, where T^{odd} counts the representations of n by $Q_{A,B}$ with all four coordinates odd; the Eisenstein part of this expansion is given by Siegel's theorem for the coset $(1+2\mathbb{Z})^4$ of $2\mathbb{Z}^4$ (Siegel's theorem with congruence conditions [15]; this is also how the genus theta series with the weights η enters in [17, p. 15]), i.e. by $M^{\mathrm{odd}}(A,B;n) = \frac{\pi^2 n}{AB}\cdot\frac12[n \equiv 2(A+B)\ (8)]\prod_{p>2}\beta_p(n)$ (for $(A,B) = (1,1)$ this is Jacobi's formula $16\sigma(n/4)$ for the number of representations of $n \equiv 4 \pmod 8$ as a sum of four odd squares). Integrating $|G|^2$ over the region above each cusp as in Lemma 6.2 gives

$$\|G_{m,m'}\|^2 \;\ge\; \Lambda(m,m') := \sum_{\delta\mid mm'}\Big[\frac{1}{4\pi}\sum_{n\ge1}\frac{|a_{G_{A,B}}(n)|^2}{n}\big(e^{-\pi\delta n/mm'} + e^{-4\pi\delta n/mm'}\big) + \frac{1}{\pi}\sum_{n\ge1}\frac{|T^{\mathrm{odd}}(A,B;n) - M^{\mathrm{odd}}(A,B;n)|^2}{n}\,e^{-\pi\delta n/mm'}\Big], \tag{18}$$

where A, B depend on δ as above. The regions above all the cusps have total area equal to the sum of the widths, which is $\mu = L\prod_{p|L}(1+1/p)$, i.e. the fraction $3/\pi = 0.955$ of $\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})$; the rest of the fundamental domain lies below height 1 at every cusp.

The script `cusp_lower_bound.py` evaluates (18). It needs the main terms $M(A,B;n)$ and $M^{\mathrm{odd}}(A,B;n)$ for *every* n, not only the admissible ones: the module `local_densities.py` computes the densities at the primes $p \mid AB$ dividing n by the recursion $\beta_p(Q_{A,B};pn') = 2(1-\frac{1}{p})[p \equiv 1\ (4)] + \frac{1}{p}\beta_p(Q_{A/p,pB};n')$ for $p \mid A$ (the non-singular solutions have $(z,w) \not\equiv 0$, the singular ones reduce to the form $Q_{A/p,pB}$), which gives $\beta_p(n) = (1+\frac{1}{p})p^{-k}$ for $p \equiv 3$ and $2-(1+\frac{1}{p})p^{-k}$ for $p \equiv 1 \pmod 4$, $k = v_p(n)$ (the recursion closes because $Q_{A/p,pB}$ is again a form of the family, with p moved to the other block, and $\beta_p = 1-\chi(p)/p$ when $p \nmid n$); the 2-adic density by sorting the solutions by the parities of (x,y) and (z,w), which gives $\beta_2(n) = \frac{1}{4}\beta_2(n/4) + \frac{1}{2}[A \equiv B\ (4)][v_2(n)=2] + [A \not\equiv B\ (4)](1+\frac{1}{2}[v_2(n)\ge 3])$ for $v_2(n) \ge 2$; and $\beta_2^{\mathrm{odd}}(n) = \frac{1}{2}[n \equiv 2(A+B)\ (8)]$. These formulae are checked exactly: at the levels 4 and 12, where there are no cusp forms, $T = M$ and $T^{\mathrm{odd}} = M^{\mathrm{odd}}$ for every $n \le 2^{17}$ to 10^{-9}, and at the level 20, $T - M = \frac{8}{3}a_{f_{20}}(n)$ for every $n \le 2^{17}$ (Section 5.2 had this only for the admissible n), while the mean square of $T^{\mathrm{odd}} - M^{\mathrm{odd}}$ grows like Y^2 as for a cusp form. With the exact main term the Rankin–Selberg estimate of Section 5 can be taken over all $n \le Y$; this is the column $\|G\|^2$ of Table 3, and for $(1,5)$ it is 0.5785, the exact value of Section 5.2.

(m,m')	mm'	$\|G\|^2$	Tab. 1	Λ_{AL}	Λ_{odd}	Λ	$\Lambda/\|G\|^2$	Thm 1.5(i)	$\frac{1}{\pi}(\log^2 m + \log^2 m')$
$(1,5)$	5	0.5785	0.74	0.496	0.046	0.543	0.938	0.496	0.82
$(3,5)$	15	2.058	3.44	1.799	0.136	1.936	0.940	1.061	1.21
$(3,7)$	21	2.481	4.02	2.186	0.167	2.352	0.948	1.183	1.59
$(5,7)$	35	4.109	4.82	3.555	0.314	3.869	0.942	2.154	2.03
$(1,101)$	101	7.429	7.95	7.147	0.100	7.247	0.976	7.147	6.78
$(13,29)$	377	12.423	13.09	10.795	1.033	11.828	0.952	6.639	5.70
$(15,77)$	1155	21.779	27.93	19.620	1.340	20.960	0.962	9.596	8.34
$(1,1009)$	1009	16.937	17.96	16.572	0.106	16.678	0.985	16.572	15.23
$(61,67)$	4087	24.704	24.47	21.411	2.000	23.411	0.948	12.875	11.01
$(101,103)$	10403	30.385	29.81	26.470	2.447	28.917	0.952	15.847	13.62
$(105,143)$	15015	47.140	49.50	40.622	3.186	43.807	0.929	17.079	14.73
$(1,15015)^*$	15015	43.233	43.54	44.610	0.230	44.839	1.037	30.818	29.44
$(1009,1013)^*$	1022117	57.730	61.23	58.545	5.271	63.816	1.105	34.469	30.47

TABLE 3. The lower bound (18) and its parts (`results/cusp_lower_bound.txt`, `_large.txt`): Λ_{AL} is the sum over the Atkin–Lehner cusps ($d_2 \in \{1,4\}$), Λ_{odd} the sum over the cusps with $d_2 = 2$, Λ their sum, "Thm 1.5(i)" the right-hand side of Theorem 1.5(i) (the four cusps with $\delta \in \{m,m'\}$), and the last column its leading term. $\|G\|^2$ is the Rankin–Selberg estimate over all $n \le 2^{21}$ with the exact main term (2^{24} for the two pairs marked *), and "Tab. 1" the estimate over the admissible n of Section 5 (for $(3,7)$, $(5,7)$, $(15,77)$, $(105,143)$ computed here in the same way). For the two pairs marked * the estimate has not converged ($Y/L = 280$ and 4; Table 1 has $S_2(Y)/S_2(Y/2) = 4.36$ and 4.56 for them), and the lower bound is the better information. By Theorem 1.5(ii), $(1,15015)$ and $(105,143)$ have the same norm, although their cusp sums differ. Floating point.

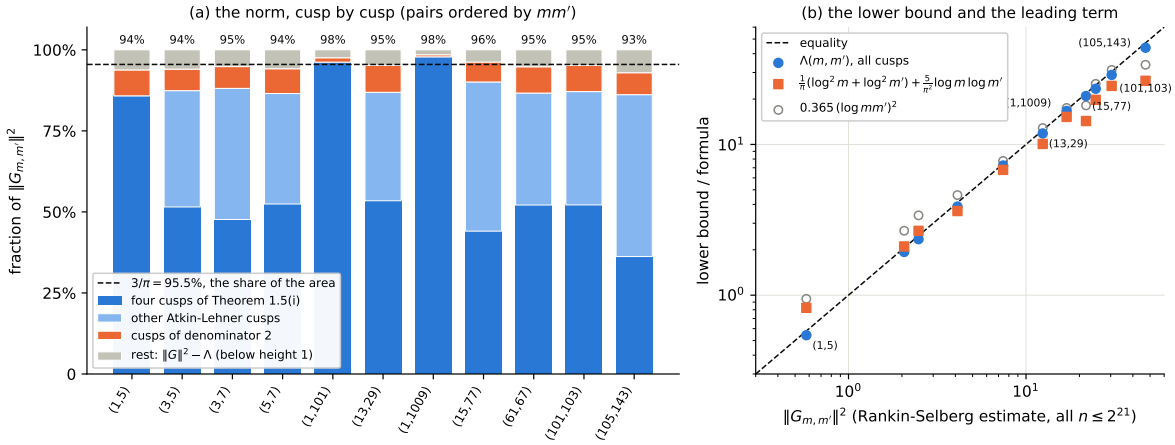


FIGURE 3. The norm decomposed over the cusps (Table 3). (a) For the eleven pairs with $Y \ge 35L$, ordered by mm', the parts of $\Lambda(m,m')$ as fractions of $\|G_{m,m'}\|^2$: the four cusps of Theorem 1.5(i), the other Atkin–Lehner cusps, the cusps of denominator 2, and the rest $\|G\|^2 - \Lambda$, which is the part of the fundamental domain below height 1; the dashed line is $3/\pi$, the share of the area above height 1, and the number above each bar is $\Lambda/\|G\|^2$. For $m = 1$ the four cusps of the theorem are all the Atkin–Lehner cusps; for $m \approx m'$ they carry about half, the other half coming from the pair (m,m') at the cusps with $\delta = 1$. (b) $\Lambda(m,m')$ and the leading term $\frac{1}{\pi}(\log^2 m + \log^2 m') + \frac{5}{\pi^2}\log m \log m'$ of the cusp sum against $\|G_{m,m'}\|^2$, with the fit $0.365(\log mm')^2$ of Section 5.3 for comparison. The cusp sums depend on the factorisation of mm', the norm does not (Theorem 1.5(ii)); $\Lambda(m,m')$ is a lower bound for the norm for every factorisation, and the pair $(1,mm')$ gives the best one. Script `plot_cusp_decomposition.py`; floating point.

Table 3 and Figure 3 show the following.

(a) For the ten pairs with $Y \ge 50L$, the regions above the cusps carry between 93.8% and 98.5% of the norm (for $(105,143)$, with $Y/L = 35$, 92.9%), about their share 95.5% of the area: the function $y^2|G|^2$ has no excess mass in the remaining 4.5% of the fundamental domain, near the elliptic points and in the "corners" between the cusps. The lower bound (18) is thus nearly an equality, and an asymptotic formula for $\|G_{m,m'}\|^2$ is a question about the lower part of the fundamental domain only.

(b) The four cusps of Theorem 1.5(i) alone give 98% for $m = 1$, where they are all the Atkin–Lehner cusps, and about half for $mm' = pq$; the rest comes from the pair (m,m') itself at the cusps with $\delta = 1$ (widths $4mm'$ and mm', and the cusp of denominator 2): for $(13,29)$, $4.16 + 1.03$ out of 12.42.

(c) For the cusps with $\delta = 1$, $T(m,m';n)$, $n \ll mm'$, is a sum of at most $1 + n/mm'$ products $r(\alpha)r(\beta)$ with $m\alpha + m'\beta = n$; replacing $r(\alpha)^2, r(\beta)^2$ by their mean values $4\log\alpha$, $4\log\beta$ (Lemma 6.3) gives for their Atkin–Lehner part

$$\frac{16}{4\pi}\iint_{u,v>0} \frac{\log(m'u)\log(mv)}{u+v}\left(e^{-\pi(u+v)} + e^{-4\pi(u+v)}\right)\mathrm{d}u\,\mathrm{d}v = \frac{5}{\pi^2}\log m \log m' + O(\log mm'),$$

since $\iint e^{-c(u+v)}(u+v)^{-1}\mathrm{d}u\,\mathrm{d}v = 1/c$: 4.38, 8.76, 10.84 against the computed 4.16, 8.54, 10.62 for $(13,29)$, $(61,67)$, $(101,103)$.

(d) The leading term $\frac{1}{\pi}((\log m)^2+(\log m')^2)$ of Theorem 1.5(i) is 85–95% of the right-hand side of (i) for $mm'\ge 100$; the difference is the term $c_1\log Y+c_2$ of Lemma 6.3 at the four cusps, which is still large at $Y\approx 10^2$–10^3.

(e) When mm' has several prime factors, the other divisors contribute: for $m=1$ and $\delta\mid m'$ the cusps with the pair $(m'/\delta,\delta)$ and width m'/δ see $T(m'/\delta,\delta;n)=r(n/\delta)$ for $n<m'/\delta$, hence contribute about $\frac{1}{\pi\delta}\bigl(\log\frac{m'}{\delta^2}\bigr)_+^2$: for $(1,15015)$ the divisors $\delta=3,5,7,11,13$ give 6.72, 2.88, 1.78, 0.82, 0.57 (against 5.84, 2.61, 1.49, 0.67, 0.49 from the formula), which is why this pair has the largest ratio $\|G\|^2/(\log mm')^2$ in Table 1.

(f) For $m=1$ and p prime the lower bound can be followed much further than the Rankin–Selberg estimate, since the four Atkin–Lehner cusps carry 98% of the norm and need only the coefficients $a_{G_{1,p}}(n)$ for $n\ll p$ (`results/cusp_lower_bound_1p.txt`):

p	101	1009	10007	100003	300007	1000003
$\Lambda(1,p)$	7.247	16.678	29.466	45.665	54.586	65.245
$\Lambda(1,p)/(\log p)^2$	0.340	0.349	0.347	0.345	0.343	0.342

The least-squares fit $\Lambda(1,p)=a(\log p)^2+b\log p+c$ over these six primes gives $a=0.319$, $b=0.42$, $c=-1.5$, against $1/\pi=0.318$ for the leading term of Theorem 1.5(ii); indeed Lemma 6.3 gives for the four cusps, before the correction by $M(1,p;n)$, exactly $\frac{1}{4\pi}\bigl(S(p/4\pi)+S(p/\pi)\bigr)=\frac{1}{\pi}(\log p)^2+0.383\log p-0.65+O(p^{-1/2})$, which is 65.39 at $p=1000003$ against the computed 65.13. The ratio $\Lambda(1,p)/(\log p)^2$ therefore tends to $1/\pi$, but only like $1/\pi+0.4/\log p$, which is why the fits of Section 5 over $mm'\le 10^4$ gave 0.37.

(g) By Theorem 1.5(ii), pairs with the same product have the same norm, although their cusp sums Λ differ (the regions above height 1 are not permuted by W_m, which changes the widths). The Rankin–Selberg estimates confirm this where they have converged (`results/norm_split_check.txt`): $(3,5)$ and $(1,15)$ give 2.0584 and 2.0591; $(3,7)$ and $(1,21)$: 2.4806 and 2.4805; $(5,7)$ and $(1,35)$: 4.1087 and 4.1067; $(13,29)$ and $(1,377)$: 12.423 and 12.390. For larger L the estimates for $(1,n)$ at $X=2^{21}$ lie below their own lower bounds $\Lambda(1,n)$ (e.g. 23.95 against 24.05 for $n=4087$, 26.66 against 29.70 for $n=10403$) and have not converged, while those for the balanced pairs have; this is also why the two rows marked * in Table 3 disagree.

In short, the cusp sums $\Lambda(m,m')$ of the fundamental domain of the pair (m,m') are, to within a few percent of the norm, $\frac{1}{\pi}((\log m)^2+(\log m')^2)+\frac{5}{\pi^2}\log m\log m'$ plus the contribution of the cusp of denominator 2 and of the other divisors of mm', plus lower-order terms: they depend on the factorisation, although their common upper bound, the norm, does not. For the pair $(1,mm')$ the leading term is $\frac{1}{\pi}(\log mm')^2$, which is the leading term of the norm itself (Theorem 1.5(iii)); its ratio to $(\log mm')^2$ approaches $1/\pi$ only like $1/\pi+0.4/\log mm'$, and the constant 0.37 of Section 5.3 is the value of this ratio in the range of the sample.

7. The two-dimensional sieve: $(7,5)$ for every N

Remark 3.3 observed that the product $n(N-n)$ is a sieve problem of dimension 2, with the level of distribution $N^{1/2-\varepsilon}$ of Part I, Theorem 3.2, and that the two-dimensional sieve of Diamond, Halberstam and Richert has the sieving limit $\beta_2=4.2665$ against $2+2\sqrt{e}=5.297$ for the vector sieve of Part I. Here we carry this out.

7.1. **The sieve functions of dimension two.** We recall the definitions [8], [7, Chapters 6 and 9]. For $\kappa\ge 1$ the Ankeny–Onishi function σ_κ is the continuous solution of

$$\sigma_\kappa(s)=\frac{s^\kappa}{(2e^\gamma)^\kappa\,\Gamma(\kappa+1)}\quad(0<s\le 2),\qquad \bigl(s^{-\kappa}\sigma_\kappa(s)\bigr)'=-\kappa s^{-\kappa-1}\sigma_\kappa(s-2)\quad(s>2),$$

and the sieve functions F_κ, f_κ are the solutions of

$$\begin{aligned} F_\kappa(s) &= \frac{1}{\sigma_\kappa(s)} \quad (0 < s \le \alpha_\kappa), & \big(s^\kappa F_\kappa(s)\big)' &= \kappa s^{\kappa-1} f_\kappa(s-1) \quad (s > \alpha_\kappa), \\ f_\kappa(s) &= 0 \quad (0 < s \le \beta_\kappa), & \big(s^\kappa f_\kappa(s)\big)' &= \kappa s^{\kappa-1} F_\kappa(s-1) \quad (s > \beta_\kappa), \end{aligned} \tag{19}$$

where $\alpha_\kappa, \beta_\kappa$ are the constants for which $F_\kappa(s), f_\kappa(s) \to 1$; $\alpha_1 = \beta_1 = 2$ and F_1, f_1 are the linear sieve functions, and $\alpha_\kappa > \beta_\kappa > 2$ for $\kappa > 1$. F_κ decreases and f_κ increases [7, Chapter 6], $F_\kappa = 1 + O(e^{-s})$, $f_\kappa = 1 - O(e^{-s})$, and both are Lipschitz: $|F_\kappa(s_1) - F_\kappa(s_2)|, |f_\kappa(s_1) - f_\kappa(s_2)| \le \kappa |s_2 - s_1|/(s_1 \sigma_\kappa(1))$ for $1 \le s_1 < s_2$ [7, Lemma 6.2]. The sieve theorem [7, Theorem 9.1], in the form in which it is quoted in [10, Section 4], states, for a sequence $\mathcal{A}$ with $|\mathcal{A}_d| = X\omega(d)/d + R_d$, ω multiplicative with $\sum_{w \le p < z} \omega(p) \log p/p \le \kappa \log(z/w) + O(1)$ (the logarithmic form of the density condition $\Omega(\kappa)$ of [7], which implies the product form used there by partial summation), $V(z) = \prod_{p<z}(1 - \omega(p)/p)$ and $2 \le z \le y$,

$$\begin{aligned} S(\mathcal{A}, z) &\le XV(z)\Big(F_\kappa\Big(\frac{\log y}{\log z}\Big) + E\Big) + 2 \sum_{\substack{m \mid P(z) \\ m < y}} 4^{\omega(m)} |R_m|, \\ S(\mathcal{A}, z) &\ge XV(z)\Big(f_\kappa\Big(\frac{\log y}{\log z}\Big) - E\Big) - 2 \sum_{\substack{m \mid P(z) \\ m < y}} 4^{\omega(m)} |R_m|, \end{aligned} \tag{20}$$

with $E = O\big((\log\log y)^2 (\log y)^{-1/(2\kappa+2)}\big)$.

The constants $\alpha_\kappa, \beta_\kappa$ are determined by two conditions, which we formulate with the adjoint functions p_κ, q_κ, the solutions of $s^\kappa p'(s) = -\kappa(s+1)^{\kappa-1} p(s+1)$ with $p(s) \sim s^{-\kappa}$ and of $s^\kappa q'(s) = \kappa(s+1)^{\kappa-1} q(s+1)$ with $q(s) \sim s^\kappa$ (Booker and Browning [2] use the equivalent $(up)' = \kappa(p(u) - p(u+1))$, $(uq)' = \kappa(q(u) + q(u+1))$). For $P = F_\kappa + f_\kappa$ and $Q = F_\kappa - f_\kappa$ the quantities

$$\begin{aligned} \Phi_P(s) &= s^\kappa P(s) p(s) + \kappa \int_{s-1}^{s} (u+1)^{\kappa-1} P(u) p(u+1)\, \mathrm{d}u, \\ \Phi_Q(s) &= s^\kappa Q(s) q(s) - \kappa \int_{s-1}^{s} (u+1)^{\kappa-1} Q(u) q(u+1)\, \mathrm{d}u \end{aligned}$$

are constant for $s \ge \max(\alpha_\kappa, \beta_\kappa) + 1$, by the differential-delay equations, and $P \to 2$, $Q \to 0$ if and only if $\Phi_P = 2$ and $\Phi_Q = 0$ (a solution Q of its equation which is not the sieve solution decays only like $s^{-2\kappa}$, and then $\Phi_Q \ne 0$). The script `dhr_sieve.py` solves (19) for given (α, β) on a grid of step $1/2000$, computes p, q by backward integration from their asymptotic series, and solves $\Phi_P = 2$, $\Phi_Q = 0$ for (α, β). For $\kappa = 1$ it returns $\alpha = \beta = 2$ and reproduces $F_1(3) = 2e^\gamma/3$, $f_1(4) = 2e^\gamma \log 3/4$ and $f_1(5/2)$ to six decimals. For $\kappa = 2$ it returns

$$\alpha_2 = 5.357727, \qquad \beta_2 = 4.266450,$$

which agree to 10^{-8} with the values $\alpha_2 = 5.35772744559\ldots$, $\beta_2 = 4.26645028414\ldots$ computed in ball arithmetic by Booker and Browning [2] (truncated at 20 decimals there), and with $\beta_2 = 4.2665$ of [8]; also $F_2(s) = 2(2e^\gamma)^2/s^2 = 25.378/s^2$ on $(0, 2]$. Refining the grid to $1/4000$ changes α_2, β_2 by $3 \cdot 10^{-8}$ and the values of F_2, f_2 below by less than 10^{-7}. Some values:

s	2	3	4	4.5	5	6	8	10
$F_2(s)$	6.3444	2.9164	1.8364	1.5704	1.3928	1.1136	1.0027	1.00002
$f_2(s)$	0	0	0	0.2403	0.5790	0.8844	0.9973	0.99998

7.2. **The weighted sieve.** We keep the setting of Part I, Section 4: $N \equiv 2 \pmod 4$ large, $\mathcal{A} = \{n \le N :\ (n,N)=1\}$ with the weights $w(n) = r(n)r(N-n)$, $X_N = \sum_{n\in\mathcal{A}} w(n)$ and, for square-free d with $(d,2N)=1$,

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n(N-n)}} w(n) = X_N \prod_{p\mid d} 2h(p) + R_d, \qquad R_d = \sum_{d_1d_2=d} R(d_1,d_2;N),$$

since $(n,N)=1$ forces $d \mid n(N-n)$ to split uniquely as $d_1 \mid n$, $d_2 \mid N-n$ (Remark 3.3); here $h(p) = (2p-1)/(p(p+1))$ for $p \equiv 1$ and $h(q) = 1/(q(q-1))$ for $q \equiv 3 \pmod 4$, as in Part I, (18). Therefore $\sum_{d\le D} d_3(d)|R_d|$ is bounded by the sum of Part I, Theorem 3.2, which is $\ll N^{1-\varepsilon/3}$ for $D = N^{1/2-\varepsilon}$, and the same holds for the sums over $d \le D/p$ needed for the sequences $\mathcal{A}_p = \{n\in\mathcal{A} :\ p \mid n(N-n)\}$. The density $\omega(p) = 2h(p)p$ satisfies $\sum_{p<z} 2h(p) = 2\log\log z + O(1)$, so the sieve has dimension $\kappa = 2$ in the sense of (20). Let $0 < a < b < \frac12$, $\lambda > 0$, $z = N^a$, $y = N^b$, and consider

$$W = \sum_{\substack{n\in\mathcal{A}\\ (n(N-n),P(z))=1}} w(n)\Big(1 - \lambda \sum_{\substack{z\le p<y\\ p\mid n(N-n)}} \Big(1 - \frac{\log p}{\log y}\Big)\Big) = S(\mathcal{A},z) - \lambda \sum_{z\le p<y} \Big(1-\frac{\log p}{\log y}\Big) S(\mathcal{A}_p, z).$$

By (20) with the lower function for $\mathcal{A}$ at the level $N^{1/2-\varepsilon}$ and the upper function for $\mathcal{A}_p$ at the level $N^{1/2-\varepsilon}/p$, and with the passage from the sum over p to an integral as in Part I, Section 4,

$$W \ \ge\ X_N V(z)\big(L_2 - \lambda R_2 - o(1)\big), \qquad L_2 = f_2\Big(\frac{1}{2a}\Big), \qquad R_2 = 2\int_a^b \Big(1-\frac{t}{b}\Big) F_2\Big(\frac{1/2-t}{a}\Big)\frac{\mathrm{d}t}{t} \tag{21}$$

(the $-\varepsilon$ in the level is absorbed by the Lipschitz continuity of f_2, F_2). On the other hand, as in Part I, Lemma 4.2, the n with $p^2 \mid n(N-n)$ for some $p \ge z$ contribute $o(X_N V(z))$, and for the others the weight is positive only if $\Omega(n(N-n)) < 1/\lambda + 2/b$ and $\max(\Omega(n),\Omega(N-n)) < 1/\lambda + 1/b$, because $\sum_{p\mid n}(1 - \log p/\log y) \ge \Omega(n) - \log n/\log y \ge \Omega(n) - 1/b$ and likewise for $N-n$, while the terms with $p \ge y$ are ≤ 0 and all prime factors are $\ge z$. Hence, exactly as in the proof of Theorem 1.1 of Part I, if $L_2 - \lambda R_2 > 0$, then for every large $N \equiv 2 \pmod 4$ the number of $n \in \mathcal{A}$ with n, $N-n$ products of distinct primes $\equiv 1 \pmod 4$ exceeding N^a, $\Omega(n(N-n)) \le k$ and $\max(\Omega(n),\Omega(N-n)) \le k'$, where $k+1 \ge 1/\lambda + 2/b$ and $k'+1 \ge 1/\lambda + 1/b$, is at least

$$\big(L_2 - \lambda R_2 - o(1)\big)\, 2^{-(k+4)} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{a^2(\log N)^2}, \tag{22}$$

with $c_{2h} = 1.66485$ as there (the factor $2^{-(k+4)}$ is $1/\max w(n)$, since $r(n) = 4\cdot 2^{\omega(n)}$ for such n).

Theorem 7.1. *For every sufficiently large $N \equiv 2 \pmod 4$,*

$$\#\big\{1 \le n < N :\ n = x^2+y^2,\ N-n = z^2+w^2,\ \Omega(n(N-n)) \le 7,\ \max(\Omega(n),\Omega(N-n)) \le 5\big\} \ge\ 0.25\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

the integers counted being coprime to N and products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{1/18}$.

Lemma 7.2.

$$f_2(9) \ \ge\ 0.9995, \qquad R_2^* := 2\int_{1/18}^{2/5} \Big(1-\frac{5t}{2}\Big) F_2(9-18t)\,\frac{\mathrm{d}t}{t} = 2\int_{1.8}^{8} \frac{s-1.8}{7.2\,(9-s)}\, F_2(s)\,\mathrm{d}s \ \le\ 2.625. \tag{23}$$

Proof. This is a computation in interval arithmetic, in the manner of Part I, Appendix A; the script is `certify_dhr.py` and its output is in the repository. Its inputs are: the enclosures

$$\alpha_2 = 5.35772744559446184227\ldots, \qquad \beta_2 = 4.26645028414864191641\ldots$$

of [2], computed there in ball arithmetic and correctly truncated at the 20th decimal (adding 10^{-20} gives upper bounds); the closed form $s^{-2}\sigma_2(s) = A_2(4 - 2\log(s/2) - 8/s + 4/s^2)$ on $[2,4]$, $A_2 = 1/(2(2e^{\gamma})^2)$, which follows from the delay equation of σ_2, and on $[4,\alpha_2]$ one more integration of that equation, enclosed cell by cell (the factor $u^{-3}(u-2)^2$ increases for $u < 6$ and $s^{-2}\sigma_2(s)$ decreases); and the monotonicity of F_2 (decreasing) and f_2 (increasing) [7, Lemma 6.2], which bounds every cell integral of (19) by its endpoint values. Closed forms are evaluated in 80-bit interval arithmetic and the recursions in IEEE doubles with an outward rounding step after each operation, on the grid of step $1/20000$. The enclosures obtained are $f_2(9) \in [0.999711,\ 0.999738]$ and $R_2^* \le 2.623903$; the floating-point values 0.99972 and 2.6238 of Section 7.1 lie inside. At $s = 4.5, 5, 6, 8$ the certified enclosures of F_2, f_2 have widths below $3 \cdot 10^{-5}$ and contain the values of the table above. □

Proof of Theorem 7.1. Take $a = 1/18$, $b = 2/5$, $\lambda = 1/3$ in (21)–(22). Then $1/\lambda + 2/b = 8$ and $1/\lambda + 1/b = 5.5$, so $k = 7$ and $k' = 5$ are admissible, $L_2 = f_2(9)$, $R_2 = R_2^*$, and $L_2 - \lambda R_2 \ge 0.9995 - 2.625/3 = 0.1245$ by Lemma 7.2; then (22) gives the constant $0.1245 \cdot 2^{-11} \cdot 8 \cdot 1.66485 \cdot 18^2 = 0.2623 > 0.25$. The remainder sums in (20) carry the weight $4^{\omega(m)}$ in place of the $d_3(de)$ of Part I, Theorem 3.2; the proof there bounds any such weight of size $N^{o(1)}$ by N^{η} (its condition (F1)), so the theorem holds with this weight as well, and the sums over $m < N^{1/2-\varepsilon}/p$ for $\mathcal{A}_p$ are sub-sums of the same sum. □

Theorem 7.1 improves the pair $(8,5)$ of Part I, Theorem 1.1, to $(7,5)$ from the same analytic input, the level $1/2$, and it is unconditional: the sieve functions enter only through Lemma 7.2, whose inputs are the certified constants of [2] and the monotonicity of F_2, f_2.

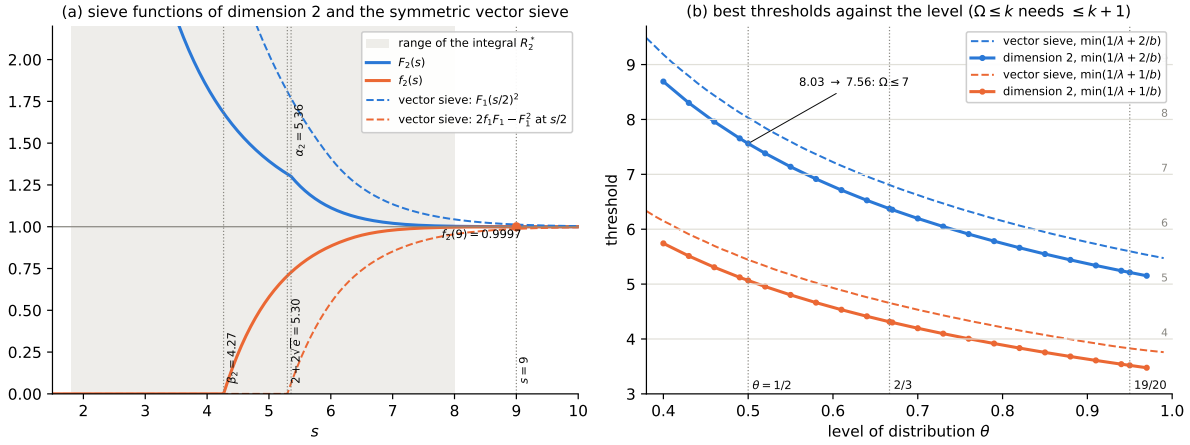


FIGURE 4. (a) The sieve functions F_2, f_2 of Section 7.1, with β_2, α_2, the point $s = 9$ of Theorem 7.1 and the range $[1.8, 8]$ of the integral R_2^* (shaded); dashed, the symmetric vector sieve of Part I for the product of two linear sieves, upper function $F_1(s/2)^2$ and lower function $2f_1(s/2)F_1(s/2) - F_1(s/2)^2$, which is positive only for $s > 2+2\sqrt{e}$. (b) The smallest attainable $1/\lambda+2/b$ and $1/\lambda+1/b$ against the level of distribution θ for the two sieves (floating point; `results/dhr_thresholds.txt` and the data of Figure 2); $\Omega(n(N-n)) \le k$, resp. $\max \Omega \le k'$, needs the value to be at most $k+1$, resp. $k'+1$. At $\theta = 1/2$ the vector sieve stops at 8.03 and the two-dimensional sieve reaches 7.56. Script `plot_dhr_sieve.py`.

7.3. **Comparison with the vector sieve.** Table 4 and Figure 4 compare, at the three levels of distribution that occur in these papers, the smallest attainable values of $1/\lambda+2/b$ and $1/\lambda+1/b$ for the two sieves (floating point; the vector sieve values are those of Figure 2), and the largest margin $L_2 - \lambda R_2$ for the pairs of interest. At the level $1/2$ the vector sieve attains 8.03 and 5.44, just missing $(7,\cdot)$, while the two-dimensional sieve attains 7.56 and 5.07: $(7,5)$ is obtained with margin 0.13, and $(7,4)$ is missed by 0.03. At $2/3$ (the level that Proposition 4.2 would give for every N from a saving $(mm')^{-1/4}$) both sieves give $(6,4)$, the two-dimensional one with margin 0.196 against 0.037; at $19/20$ the two-dimensional sieve gives $(5,3)$ with margin 0.24, where the vector sieve reached $(5,3)$ only at the edge of its parameter range (Section 3).

θ	vector sieve		dimension 2		margin $L_2 - \lambda R_2$ for the pair, at (a,b,λ)	
	$1/\lambda+2/b$	$1/\lambda+1/b$	$1/\lambda+2/b$	$1/\lambda+1/b$		
1/2	8.03	5.44	7.56	5.07	$(7,5)$: $+0.132$ (0.054, 0.42, 0.309)	$(7,4)$: -0.030
2/3	6.79	4.64	6.37	4.50	$(6,4)$: $+0.196$ (0.076, 0.50, 1/3)	$(5,4)$: -0.157
19/20	5.60	3.83	5.21	3.75	$(5,3)$: $+0.238$ (0.124, 0.50, 1/2)	$(4,3)$: -0.099

TABLE 4. The two sieves for $n(N-n)$ (`results/dhr_sieve.txt`, `results/sieve_thresholds.txt`): the bound $\Omega(n(N-n)) \le k$, resp. $\max \Omega \le k'$, requires $1/\lambda + 2/b \le k+1$, resp. $1/\lambda + 1/b \le k'+1$, with $L - \lambda R > 0$. The vector sieve values at $2/3$ are those of Figure 2 at $\theta = 0.67$. Floating point.

8. BEYOND THE LEVEL 1/2: THE SUM OVER THE MODULI

We end with the question that the sieve route leaves open for every N: the level of distribution beyond $1/2$. At the level θ, the remainder of the sieve (the vector sieve of Part I as well as the

two-dimensional sieve of Section 7) is a sum

$$\sum_{\substack{de\le N^{\theta}\\ (d,e)=1,\ (de,2N)=1}} \lambda_d\lambda'_e\, R(d,e;N) \tag{24}$$

with sieve weights $|\lambda_d|, |\lambda'_e| \le 1$ supported on square-free integers, and by (4) each $R(d,e;N)$ is a bounded combination of coefficients $a_{G_{d',e'}}(N')$ with $d'e' \asymp de$ (up to divisors of N) and $N' \in \{N, N/2\}$. Part I bounds (24) by $\sum |R(d,e;N)|$ and each term by $N^{1/2+\varepsilon}$ (Theorem 1.1(ii)), which gives $N^{1/2+\varepsilon}N^{\theta}$ and hence $\theta < 1/2$; Proposition 4.2 shows that any power saving $(de)^{-\eta}$ in the individual bound raises the level. The other possibility is cancellation between the terms: $R(d,e;N)$ changes sign with (d,e), and the sieve weights are sign-changing too.

8.1. **The structure of the sum.** Since $\sum_{d|n,\ e|N-n} r(n)r(N-n) = T(d,e;N)$ for coprime d, e, the sum (24) is, before the Möbius inversion of $(n,N)=1$,

$$\sum_{d,e} \lambda_d\lambda'_e\big(T(d,e;N) - M(d,e;N)\big) = \sum_{d,e} \lambda_d\lambda'_e\Big(\sum_{d\alpha+e\beta=N} r(\alpha)r(\beta) - M(d,e;N)\Big),$$

a weighted count of the representations of N by the family of quaternary forms $Q_{d,e}$. Writing $n = d\alpha$ and $N - n = e\beta$, the main sum is

$$\sum_{e} \lambda'_e \sum_{\substack{d,\alpha\\ d\alpha\equiv N\ (e)}} \lambda_d\, r(\alpha)\, r\Big(\frac{N-d\alpha}{e}\Big), \tag{25}$$

the distribution of the products $d\alpha$, weighted by $\lambda_d r(\alpha)$, in the residue class N modulo e, with the additional weight $r((N-d\alpha)/e)$ attached to the quotient. Without the last factor this is the setting of the dispersion method of Bombieri, Friedlander and Iwaniec [3]: sums $\sum_q \lambda_q \sum_{mn\equiv a\ (q)} \alpha_m\beta_n$ with well-factorable λ, for which the level $4/7$ is known for suitable α, β from the Deshouillers–Iwaniec bounds for sums of Kloosterman sums. The factor $r((N-d\alpha)/e) = 4\sum_{k|(N-d\alpha)/e} \chi(k)$ introduces a third modulus: (25) becomes $4\sum_{e,k} \lambda'_e\chi(k) \sum_{d\alpha\equiv N\ (ek)} \lambda_d r(\alpha)$, the distribution of $d\alpha$ in progressions to the moduli ek, where k runs up to $N/(de)$, so that the moduli ek range up to N. This is the shape of the Titchmarsh divisor problem, $\sum_q \sum_{mn\equiv a\ (q)}$ over all $q \le x$, which the dispersion method does handle (Fouvry [9], Bombieri–Friedlander–Iwaniec [3]), with the difference that here both $d\alpha$ and the quotient carry the arithmetic weight r and the modulus e carries a sieve weight. We expect that the dispersion method, applied to (25) with the moduli e in a range N^{θ_e} and the bilinear structure of $\lambda_d r(\alpha)$, gives a level $\theta > 1/2$ for the signed sum (24) and with it an improvement of Theorem 7.1 for every N; we have not carried this out. The spectral version of the same question is the sum over the levels $4de$ of the inner products $\langle G_{d,e}, P_N^{(4de)}\rangle$ of Section 4.

8.2. **The signs of $a_{G_{d,e}}(N)$, numerically.** Whether cancellation of this kind is present can be measured. The script `modulus_sum_scan.py` computes, for a fixed $N \equiv 2 \pmod 4$ and all coprime odd square-free pairs (d,e) with $de \le N^{3/4}$, the coefficients $a_{d,e} = a_{G_{d,e}}(N) = T(d,e;N) - M(d,e;N)$ exactly (the main term for every N by the formulae of Section 6.6), and compares the sums

$$A = \sum |a_{d,e}|, \qquad Q = \Big(\sum a_{d,e}^2\Big)^{1/2},$$
$$S_1 = \sum a_{d,e}, \qquad S_{\mu} = \sum \mu(de)\, a_{d,e}, \qquad S_{\mathrm{rnd}} = \sum \pm a_{d,e}$$

(all sums over the pairs with $de \le N^\theta$), where $\pm$ are independent random signs; $|S|/Q \approx 1$ is square-root cancellation and A/Q is the maximum. Table 5 shows the result.

N	θ	pairs	A/N	Q/N	S_1/Q	S_μ/Q	$S_{\rm rnd}/Q$	A/Q
$2000002 = 2 \cdot 101 \cdot 9901$	0.50	2149	0.82	0.024	−0.43	0.66	1.93	35
	0.60	10367	2.43	0.033	−0.75	0.55	−0.05	73
	0.70	49699	6.78	0.044	1.43	−0.76	0.39	154
	0.75	108127	10.93	0.049	2.01	−1.30	0.69	221
$2000006 = 2 \cdot 1000003$	0.50	2149	0.85	0.025	2.15	0.38	0.29	35
	0.60	10367	2.49	0.034	2.34	−0.11	1.50	74
	0.70	49699	6.83	0.044	0.40	−0.20	1.82	154
	0.75	108131	11.01	0.050	1.74	−0.75	2.88	221
$2000018 = 2 \cdot 293 \cdot 3413$	0.50	2149	0.86	0.025	−1.87	−1.34	0.63	34
	0.60	10367	2.49	0.034	−1.78	−0.54	1.09	73
	0.70	49699	6.77	0.045	−1.13	1.04	−0.08	152
	0.75	108131	10.90	0.050	0.71	1.29	0.89	218
$3000002 = 2 \cdot 557 \cdot 2693$	0.50	2677	0.78	0.021	0.51	0.37	1.08	38
	0.60	13515	2.35	0.029	1.05	−0.25	0.72	82
	0.70	67357	6.63	0.038	2.18	−0.97	1.52	177
	0.75	149721	10.81	0.042	3.28	−0.06	1.08	257

TABLE 5. Sums of $a_{G_{d,e}}(N)$ over the coprime odd square-free pairs with $de \le N^\theta$ (`results/modulus_sum_scan.txt`). Floating point.

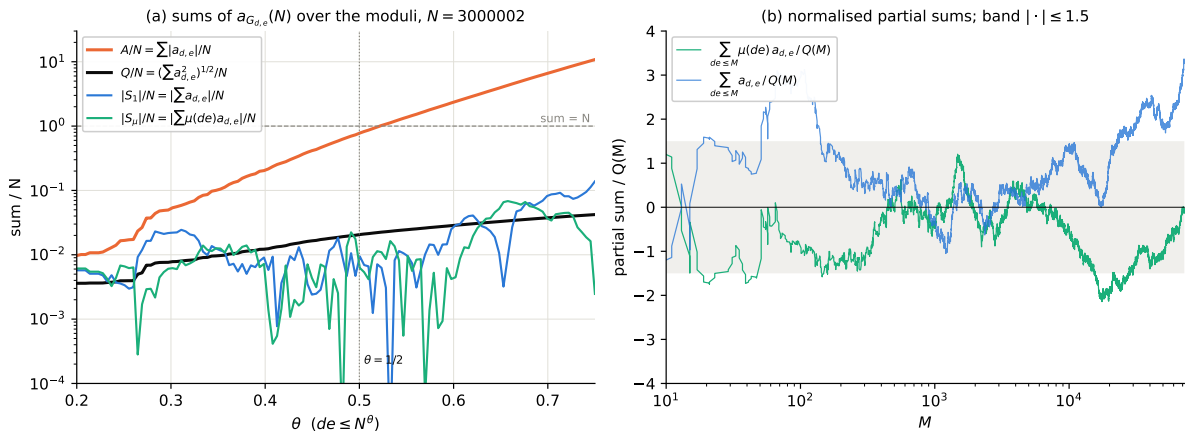


FIGURE 5. Cancellation in the sum over the moduli for $N = 3000002$ (Table 5). (a) A/N, Q/N, $|S_1|/N$ and $|S_\mu|/N$ as functions of θ, the sums running over the coprime odd square-free pairs with $de \le N^\theta$: the sum of absolute values passes N near $\theta = 0.55$, while the signed sums stay at the level $Q/N \approx 0.02$–0.05 of square-root cancellation. (b) The partial sums $\sum_{de\le M} \mu(de) a_{d,e}$ and $\sum_{de\le M} a_{d,e}$ divided by $Q(M) = (\sum_{de\le M} a_{d,e}^2)^{1/2}$, against M; the band is $|\cdot| \le 1.5$. Script `plot_modulus_sums.py`; floating point.

For all three weightings the sums are of the size Q, the square root of the sum of squares, as for independent random signs ($|S_\mu|/Q \le 1.3$, $|S_{\rm rnd}|/Q \le 2.9$, $|S_1|/Q \le 3.3$ in the table): the coefficients $a_{G_{d,e}}(N)$ for different (d, e) show no sign correlation with each other or with $\mu(de)$, and the signed sums are smaller than $\sum |a_{d,e}|$ by the factor A/Q, which grows with θ

(from 35 at $\theta = 1/2$ to 220 at $3/4$). The sum A itself is already larger than N for $\theta \geq 0.55$ at these N, which is why the proofs need N large; the signed sums are 0.02–$0.05\,N$. Since $|a_{d,e}| \approx N^{1/2}(de)^{-1/2}\log(de)$ on average (Theorem 1.2 and Section 5), $Q \approx N^{1/2}(\log N)^{3/2}$ for every θ, while $A \approx N^{1/2+\theta/2}$; both are $o(N)$ for $\theta < 1$ if Conjecture 4.1 holds, but only the cancellation visible here is available without it. The data are, of course, no substitute for a proof: they say that the sum (24) behaves as the dispersion method would predict, not how to prove it.

Data and code

The scripts `mean_square_check.py`, `norm_level20.py`, `certify_modular.py` (arguments `1/2`, `2/3`, `19/20`), `plot_mean_square.py`, `plot_sieve_thresholds.py`, `plot_cusp_decomposition.py`, `plot_dhr_sieve.py` and `plot_modulus_sums.py` (Figures 1, 2, 3, 4, 5), `local_densities.py` and `cusp_lower_bound.py` (Table 3), `dhr_sieve.py` (Section 7, output `results/dhr_sieve.txt`), `certify_dhr.py` (Lemma 7.2, output `results/certify_dhr_output.txt`), `modulus_sum_scan.py` (Table 5, output `results/modulus_sum_scan.txt`), `test_rs_formula.py` and `test_rs_old.py` (Remark 6.9, outputs `results/test_rs_formula.txt`, `results/test_rs_old.txt`), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/norm_level20.txt`, `results/certify_modular_output_twothirds.txt`, `results/certify_modular_output_nineteen.txt`, `results/sieve_thresholds.txt`, `results/cusp_lower_bound.txt`, `results/cusp_lower_bound_large.txt`, `results/cusp_lower_bound_1p.txt`, `results/norm_split_check.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the computations of Section 5 are in floating point and are evidence, not proof.

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